

LM/IPC620 IO to MLPLC IO Wiring Kit User's Guide

WK21-200
Release Independent
March 2013

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Honeywell Process Solutions
1860 W. Rose Garden Lane
Phoenix, AZ 85027 USA
1-800 822-7673

About This Document

This document provides information about the replacement of the Logic Manager (LM) I/O using the Wiring Kit. It includes procedures for installing the wiring kit which facilitates the conversion of LM I/O to the Master Logic Programmable Logic Controller (ML PLC) I/O. In addition, it discusses about the following:

- Wiring Kit and its components
- Service and maintenance

Release Information

Document Name	Release Number	Publication Date
LM/IPC620 IO to MLPLC IO Wiring Kit User's Guide - WK21-200	Release Independent	March 2013
Document Category Installation and Upgrades		

References

The following list identifies documents that are sources of information for certain topics in this publication.

Document Title	Doc ID
TPS/TPN Knowledge Builder	
• Logic Manager Installation	LM20-500
• Logic Manager Service	LM13-500
• Data Entity Builder Manual	SW11-611
• Documentation Tool Manual	SW11-609
• Process Manager I/O Installation (PM/APM/HPM) Manual	PM20-620
MLPLC User documentation	
• Programmable Logic Controller MasterLogic-200R User's Guide	Not applicable

Intended users

This guide is intended for the following users.

- Users of TPN migrating from LM to MLPLC
- Technical Assistance Centre
- Project engineering team

Prerequisite skills

The information provided in this guide is based on the assumption that you have a basic knowledge of the following:

- LM operations
- MLPLC configuration and operation
- Engineering conventions/practices of the Plant

How to use this guide

The information in this guide is arranged based on the tasks you may want to perform, as listed in the following table.

If you want to	Refer to
Understand the background and benefits of replacing the Logic Manager with ML PLC	<i>Introduction to LM migration</i> section
Plan for the LM replacement	<i>Planning for LM replacement</i> section
Understand the prerequisites for the LM replacement	<i>Replacing the LM processor</i> section
Perform the Step-by-step procedure for the LM replacement	• <i>Replacing the LM processor</i> section • <i>Replacing the LM I/O rack</i> section • <i>Post-migration checks</i> section
Service and maintain the Wiring kit	<i>Service and maintenance</i> section

Acronyms and Definitions

Acronym	Definition
ACM	ASCII Communication Module
AM	Application Module
CL	Control Language
EB	Exception Build
ESD	Electro Static Discharge
FTA	Field Termination Assembly
LCS	Logic Control System
LM	Logic Manager
LU	Link Unit
MLPLC	Master Logic Programmable Logic Controller
PCB	Printed Circuit Board
RTD	Resistance Temperature Detector
RTP	Remote Terminal Panel
TB	Terminal Block
TPN	TotalPlant® Network
TPS	TotalPlant® Solution
UAIM	Universal Analog Input Module
UCN	Universal Control Network

Agency Compliance Information

This product complies with:



- EN 61131-2: 2007



- C22.2 No.1010.1

Support and Other Contacts

United States and Canada

Contact: Honeywell Solution Support Center
Phone: 1-800-822-7673
Calls are answered by dispatcher between 6:00 am and 4:00 pm Mountain Standard Time. Emergency calls outside normal working hours are received by an answering service and returned within one hour.
Fascimile: 1-973-455-5000
Mail: Honeywell TAC, MS L17
1860 W. Rose Garden Lane
Phoenix, AZ, 85027 USA

Europe, Middle East, and Africa (EMEA)

Contact: Honeywell TAC – EMEA
Phone: +32-2-728-2345
Fascimile: +32-2-728-2696
Mail: TAC-BE02
Hermes Plaza
Hermeslaan, 1H
B-1831 Diegem, Belgium
Email: tac-be02@honeywell.com

Pacific

Contact: Honeywell Global TAC – Pacific
Phone: 1300-364-822 (toll free within Australia)
+61-8-9362-9559 (outside Australia)
Fascimile: +61-8-9362-9564
Mail: Honeywell Limited Australia
5 Kitchener Way
Burswood, WA, 6100, Australia
Email: GTAC@honeywell.com

India

Contact: Honeywell Global TAC – India
Phone: +91-20- 6603-2718/19
1800-233-5051
Fascimile: +91-20- 6603-9800
Mail: Honeywell Automation India Ltd
56 and 57, Hadapsar Industrial Estate
Hadapsar, Pune –411 013, India
Email: Global-TAC-India@honeywell.com

Korea

Contact: Honeywell Global TAC – Korea
Phone: +82-80-782-2255 (toll free within Korea)
Fascimile: +82-2-792-9015
Mail: Honeywell Co., Ltd
4F, Sangam IT Tower B4-4 Block
1590, DMC Sangam-dong, Mapo-gu
Seoul, 121-835, Korea
Email: Global-TAC-Korea@honeywell.com

People's Republic of China

Contact: Honeywell Global TAC – China
Phone: +86- 21-2219-6888
800-820-0237
400-820-0386
Mail: Honeywell (China) Co., Ltd
33/F, Tower A, City Center, 100 Zunyi Rd.
Shanghai 200051, People's Republic of China
Email: Global-TAC-China@honeywell.com

Singapore

Contact: Honeywell Global TAC – South East Asia
Phone: +65-6823-2215
Fascimile: +65-6445-3033
Mail: Honeywell Private Limited
Honeywell Building
17, Changi Business Park Central 1
Singapore 486073
Email: GTAC-SEA@honeywell.com

Japan

Contact: Honeywell Global TAC – Japan
Facsimile: +81-3-6730-7228
Mail: Honeywell K.K.
New Pier Takeshiba, South Tower Building
20th Floor, 1-16-1 Kaigan, Minato-ku
Tokyo 105-0022, Japan
Email: Global-TAC-JapanJA25@honeywell.com

Elsewhere

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	CAUTION: Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation, which, if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.

Symbol Definitions

Symbol	Definition
	WARNING, Risk of electrical shock: Potential shock hazard where HAZARDOUS LIVE voltages greater than 30 Vrms, 42.4 Vpeak, or 60 VDC may be accessible.
	ESD HAZARD: Danger of an electro-static discharge to which equipment may be sensitive. Observe precautions for handling electrostatic sensitive devices.

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1. Introduction

1.1 Overview of LM migration

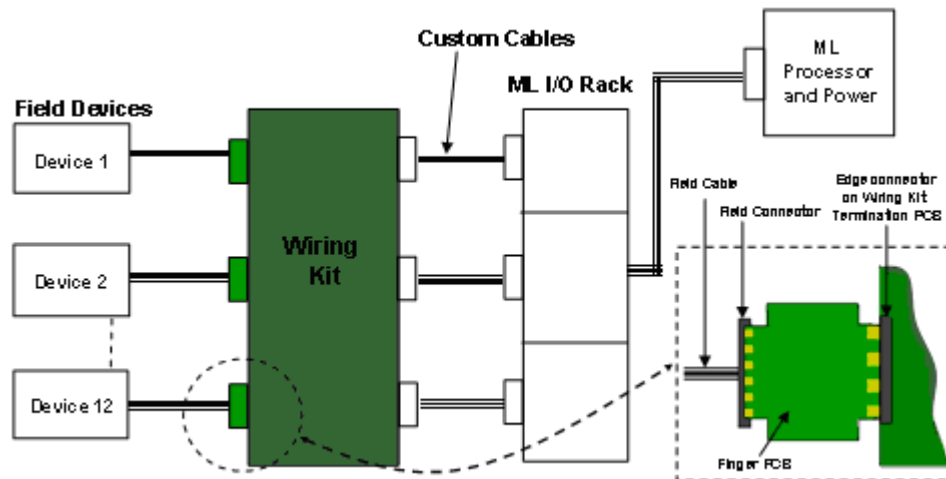
Background

Logic Manager Controller (LM) is replaced by the next generation Programmable Logic Controller (PLC) called Master Logic Programmable Logic Controller (MLPLC). With the LM to MLPLC migration, you can use the enhanced features provided by MLPLC. The MLPLC can successfully replace the existing LM controllers with its enhanced features.

Scope

This document applies to all existing LM implementations that are migrated to the MLPLC. The most beneficial feature of this migration solution is the use of the Wiring Kit. The Wiring kit is available only for LM to MLPLC migration.

The Wiring Kit is an interface between the existing field I/O termination and the MLPLC I/O rack. Wiring Kit helps in avoiding the field re-wiring, as illustrated in the following figure.



1 Introduction

1.2 Stages in migration



REFERENCE - INTERNAL

For more information about the Wiring Kit, see the *Wiring Kit solution* section.

1.2 Stages in migration

The following table identifies the stages in the migration process.

Stages	For more information, see...
Knowing Wiring Kit as a solution to migration	<i>Wiring Kit Solution</i> section
Planning for the LM replacement	<i>Planning for LM Replacement</i> section
Understanding the pre-requisites for LM replacement	<i>Replacing the LM processor</i> topic
Migrating LM database	<i>Contact the Honeywell Team</i>
Migrating LM hardware	<i>Replacing the LM I/O rack</i> section

Hardware migration

The associated hardware consists of MLPLC, a Wiring Kit interface to MLPLC, a +24 volts DC Wiring Kit power supply, and the existing LM rack with all its components. The migration is performed to replace the existing LM processor with the MLPLC processor and the LM I/O rack/s with the Wiring Kit and MLPLC I/O rack/s.

Software migration

The software migration includes the conversion of the existing LM database to the MLPLC database.

1.3 Wiring Kit solution

Benefits

The Wiring Kit performs an intermediary function between LM I/O connectors and MLPLC I/O connectors. It retains the existing LM I/O connectors connecting to the field devices on one side and connects to the MLPLC I/O connectors on the other end. The Wiring Kit facilitates the use of all the MLPLC features.

The following are the advantages of using the Wiring Kit.

- Field wiring is not changed during the LM to MLPLC migration. The elimination of rewiring reduces the time taken for migration, labor, and possibility of human error, thereby reducing the migration costs.
- The Wiring Kit considers the LCS I/O variations and maps them to the MLPLC I/O, resulting in optimum signal match solution. This reduces the installation time in the field.
- Common solution for Direct wired and Remote Terminal Panel (RTP) types.
- Channel isolation feature of LM is not available in MLPLC. However, this feature is made available in MLPLC by using the Wiring Kit.
- Additional panel space is typically not required. The Wiring Kit can reuse the existing LM panel in many cases.
- Installation of MLPLC in a different panel other than the existing LM is supported.
- The existing +24 volts DC power supply from the HPM panel can be used if the required current for wiring kit is available.

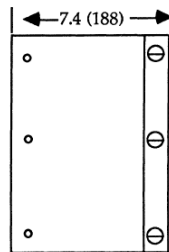
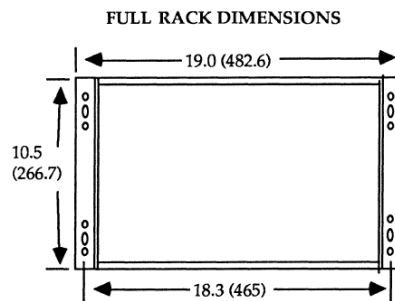
1 Introduction

1.3 Wiring Kit solution

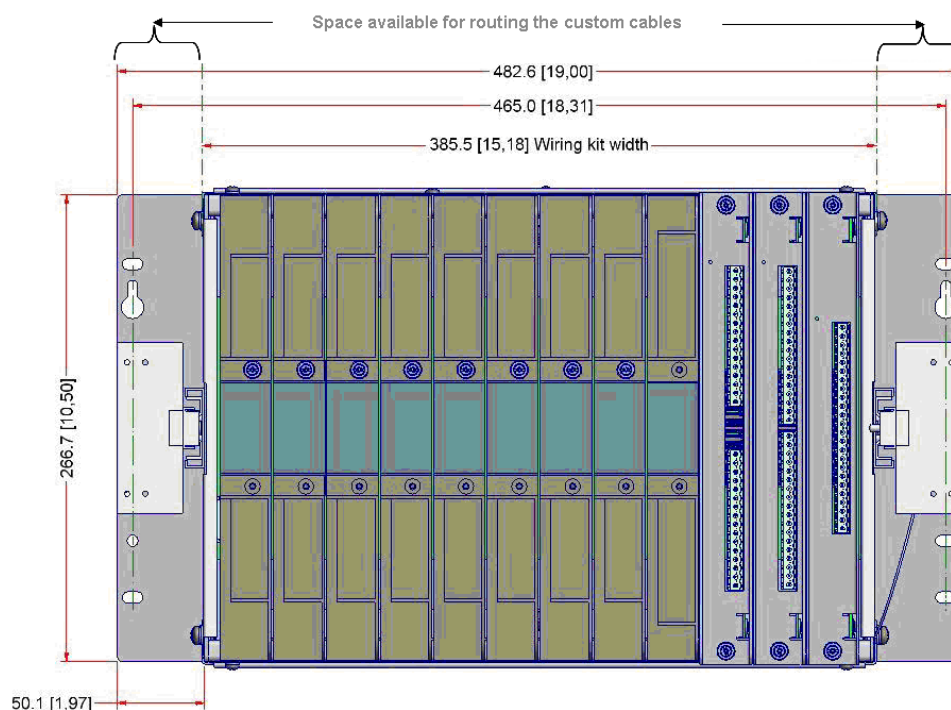
Wiring Kit dimensions

The Wiring Kit can be easily installed in the I/O rack similar to installing any other I/O rack. The Wiring Kit exactly fits into the same location of the replaced LM.

The following figures illustrate the dimensions of a full rack.



Dimensions are inches (millimeters)



The following table compares the size of the Wiring Kit (full rack) with the replaced LM.

Size along	Size (millimeters)		Difference
	LM	Wiring Kit with MLPLC	
Width	19.0 (482.6)	19.0 (482.6)	Nil
Length	10.5 (266.7)	10.5 (266.7)	Nil
Depth	7.4 (188)	12 (301.5)	4.6 (113.5)

The cabinets housing LM I/Os are reused for housing the wiring kit racks. Wiring Kit is identical to Card file assembly of LM I/Os.

1 Introduction

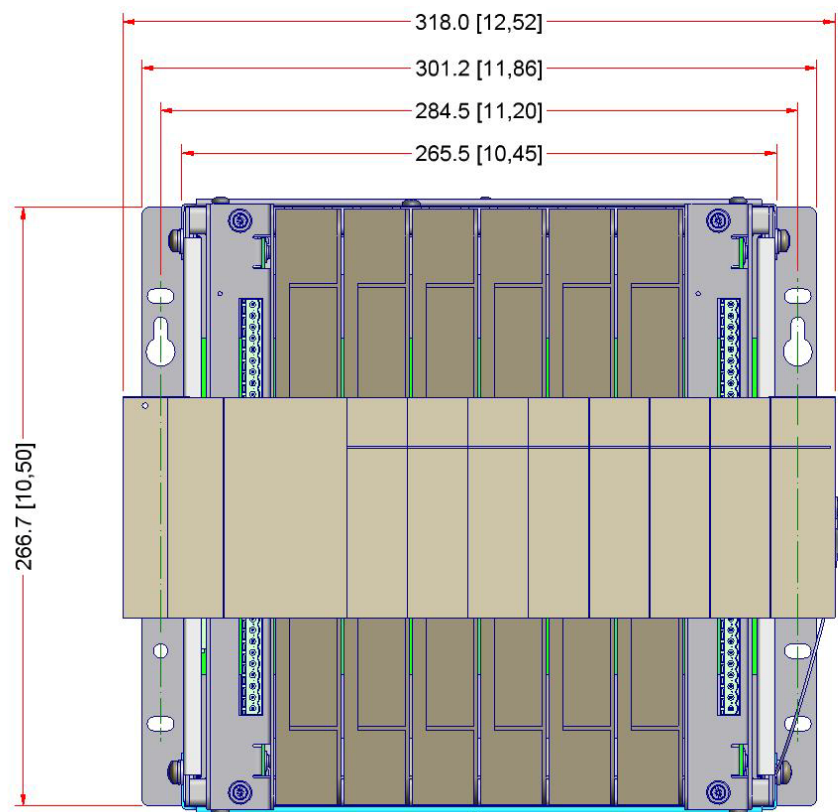
1.3 Wiring Kit solution



REFERENCE - INTERNAL

For more information about Cabinet, see the *Card File Installation* section in the *Logic Manager Installation*.

The following figure illustrates the dimensions of a half rack.

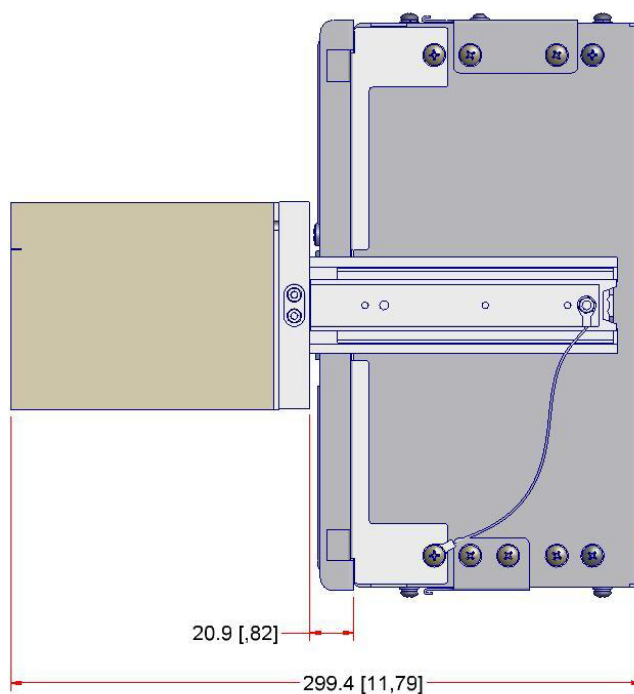


The following table compares the size of the Wiring Kit (half rack) with the replaced LM.

Size along	Size (millimeters)		Difference
	LM	Wiring Kit with MLPLC	
Width	11.8 (301.2)	13.7 (347.9)	1.9 (46.7)
Length	10.5 (266.7)	10.5 (266.7)	Nil
Depth	7.4 (188)	12 (301.5)	4.6 (113.5)

The following figures illustrate the dimensions of a half rack and a full rack indicating the depth.

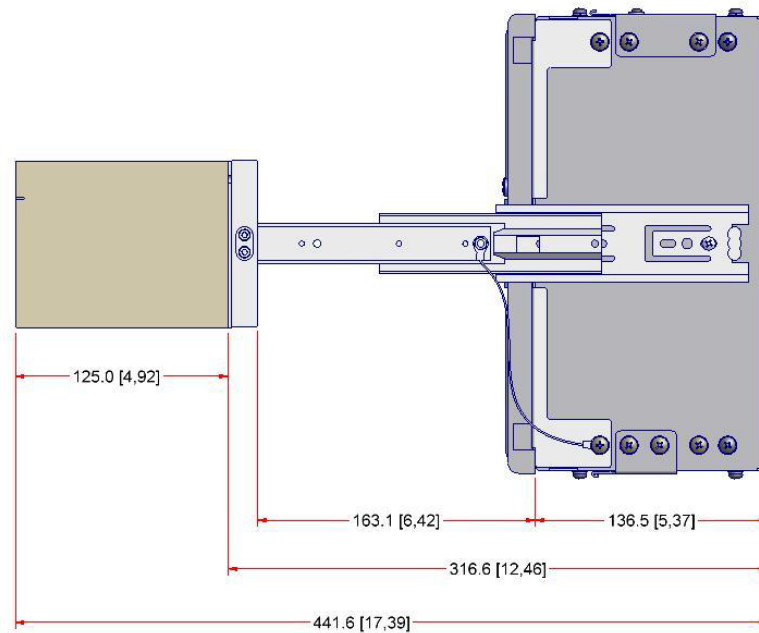
ML PLC Retracted



1 Introduction

1.3 Wiring Kit solution

ML PLC Extended



As the width of the Wiring Kit is less than the I/O rack width, additional space is available adjacent to the Wiring Kit. If the MLPLC I/O module is mounted above the Wiring Kit, the additional space can be used for routing the custom cables from the Wiring Kit to the MLPLC I/O module.

As the size of the MLPLC processor is smaller, additional space is available above the I/O rack. This space can accommodate cable terminations from the top. If the MLPLC is mounted on a different rack, the available additional space can be used for routing the custom cables amidst the already existing cables.

Full Cabinet

The following figure illustrates the full cabinet view.



1 Introduction

1.4 Wiring Kit components

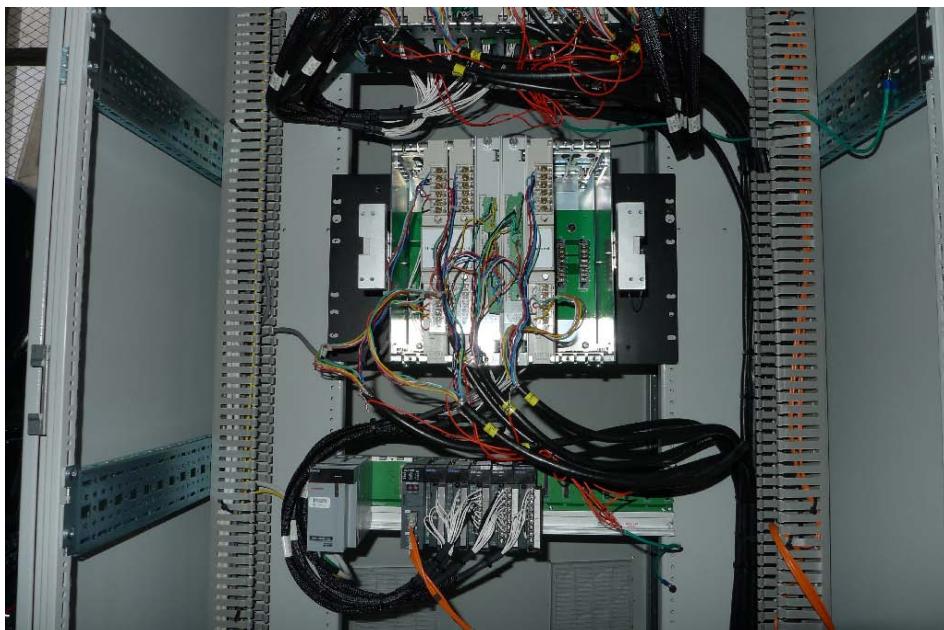
Full Rack

The following figure illustrates the full rack view.



Half Rack

The following figure illustrates the half rack view.



1.4 Wiring Kit components

Overview

This section discusses the various Wiring Kit components involved in the LM migration solution.

- Finger PCB (Printed Circuit Board)
- Custom cables
- Retractable I/O base
- +24V DC Wiring Kit power supply (Required only if an Active Finger PCB is installed in the Wiring Kit)
- Power supply accessories
 - Power supply bracket
 - Distribution board
 - Cables



ATTENTION

The +24V DC power supply is not part of the Wiring Kit and must be ordered separately. In addition, the accessories needed for mounting the power supply must be ordered separately.

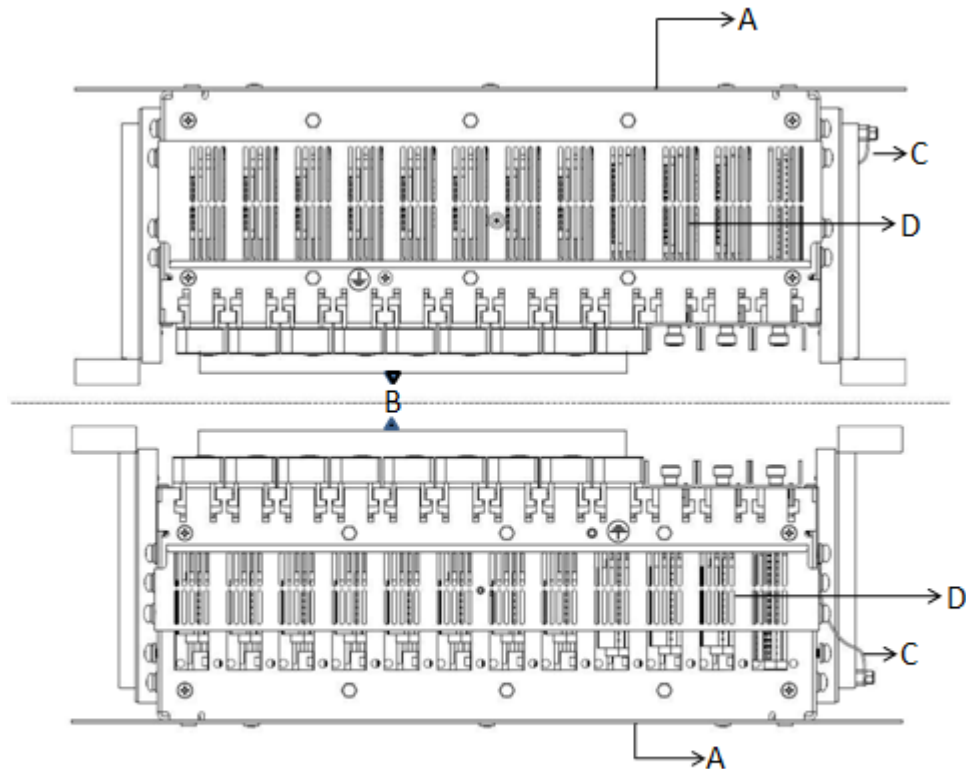
1 Introduction

1.4 Wiring Kit components

Wiring Kit

The following drawings illustrate the different views of the Wiring Kit.

Top and bottom view



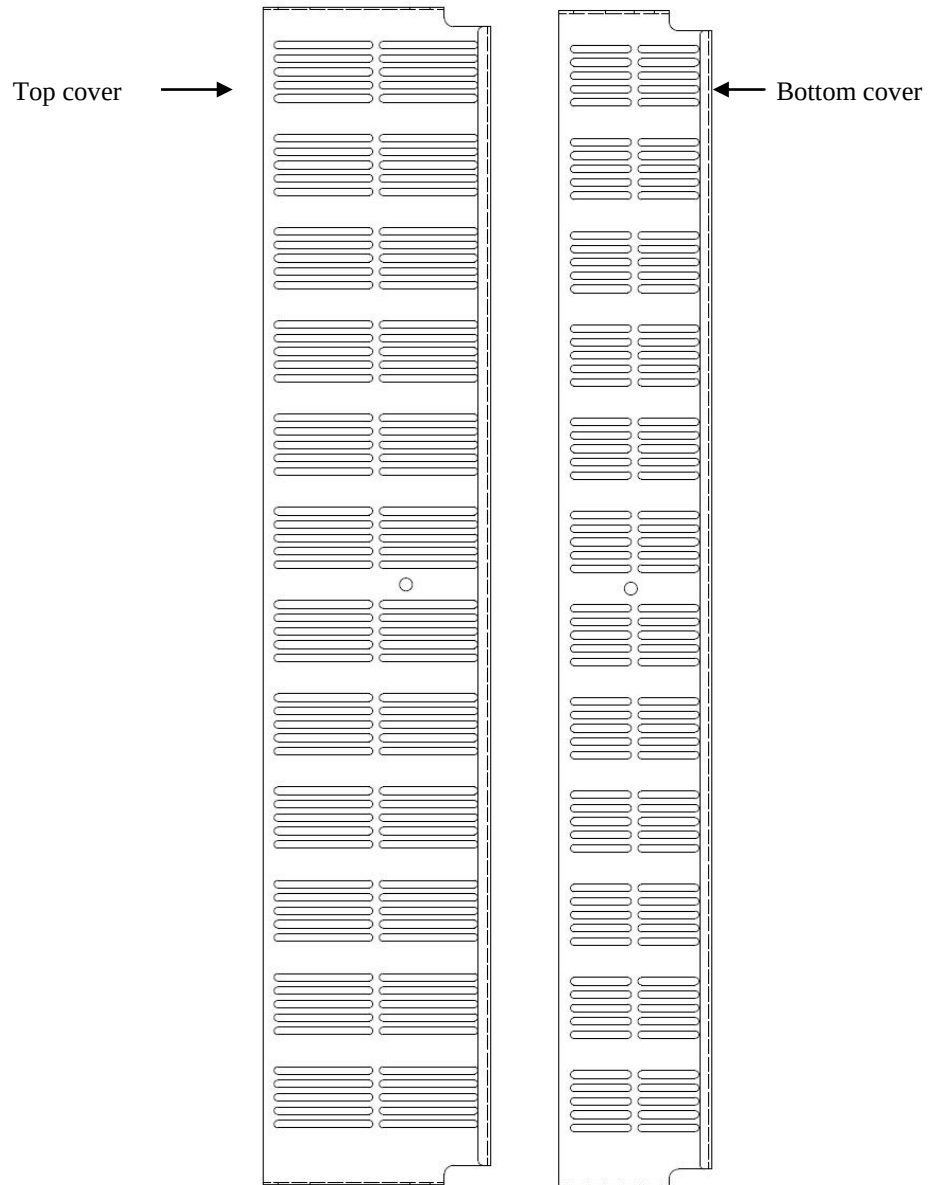
- A – Back mounted on the panel or wall
- B – Swing arm connectors connected to the field wiring in the front
- C – Locking screws
- D – ESD cover on top and bottom of the chassis

The ESD cover present on top and bottom of the chassis is protective equipment.

Ensure to keep this closed whenever the wiring kit is powered up and active.

Full Rack

The following figure illustrates the full rack view of the ESD cover.

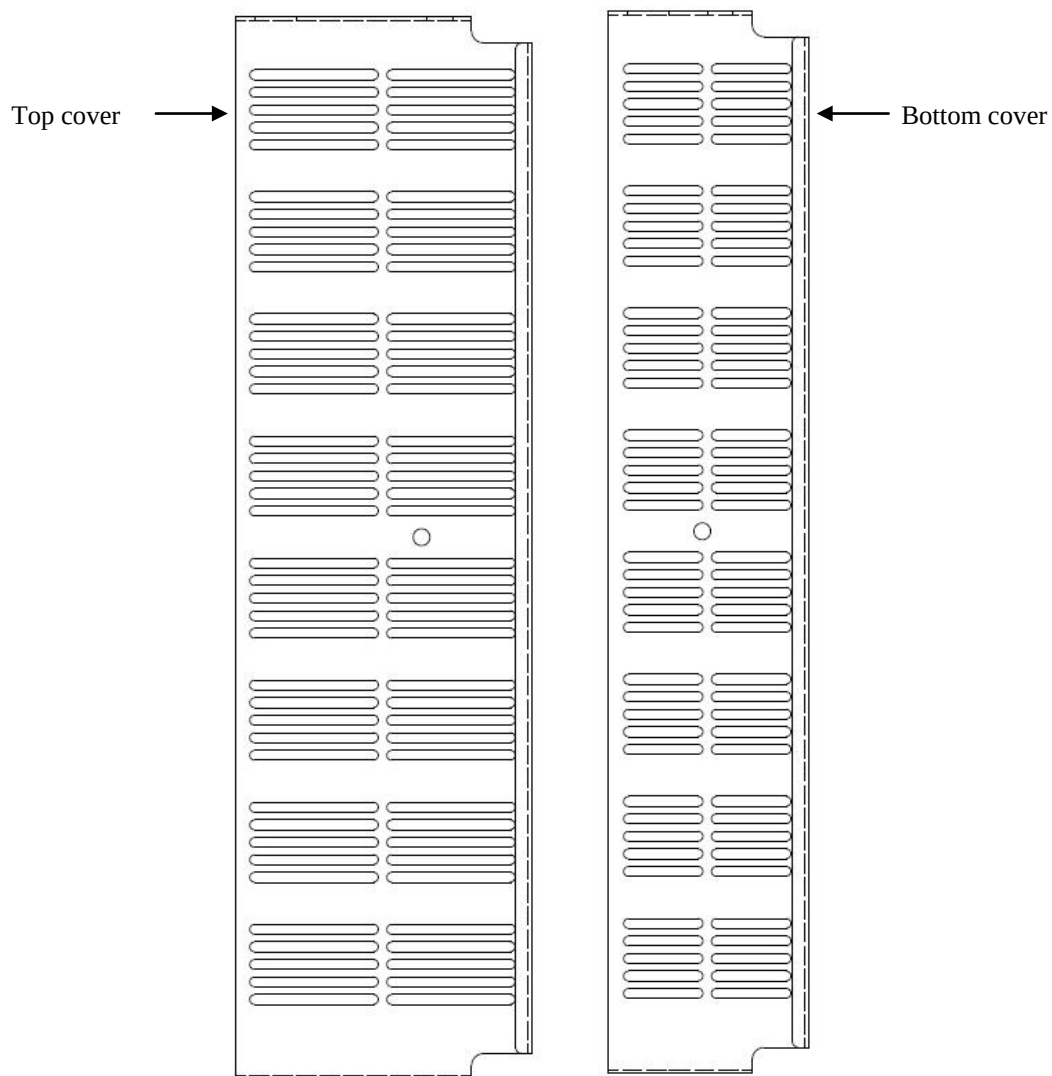


1 Introduction

1.4 Wiring Kit components

Half Rack

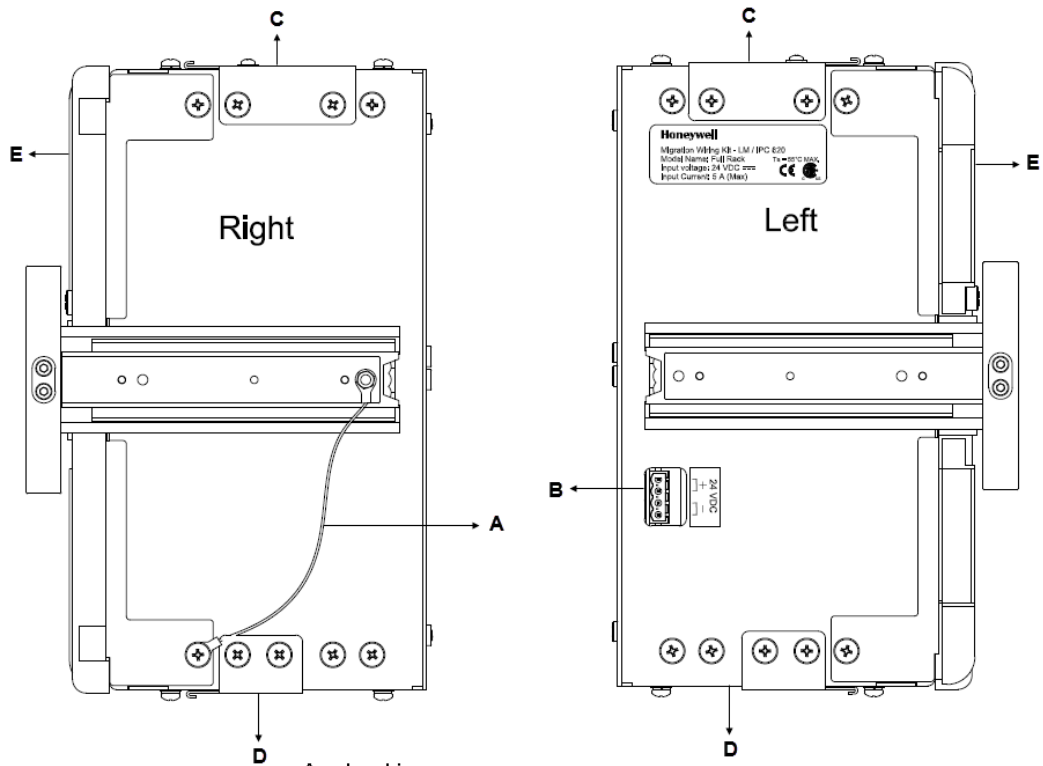
The following figure illustrates the half rack view of the ESD cover.



1 Introduction

1.4 Wiring Kit components

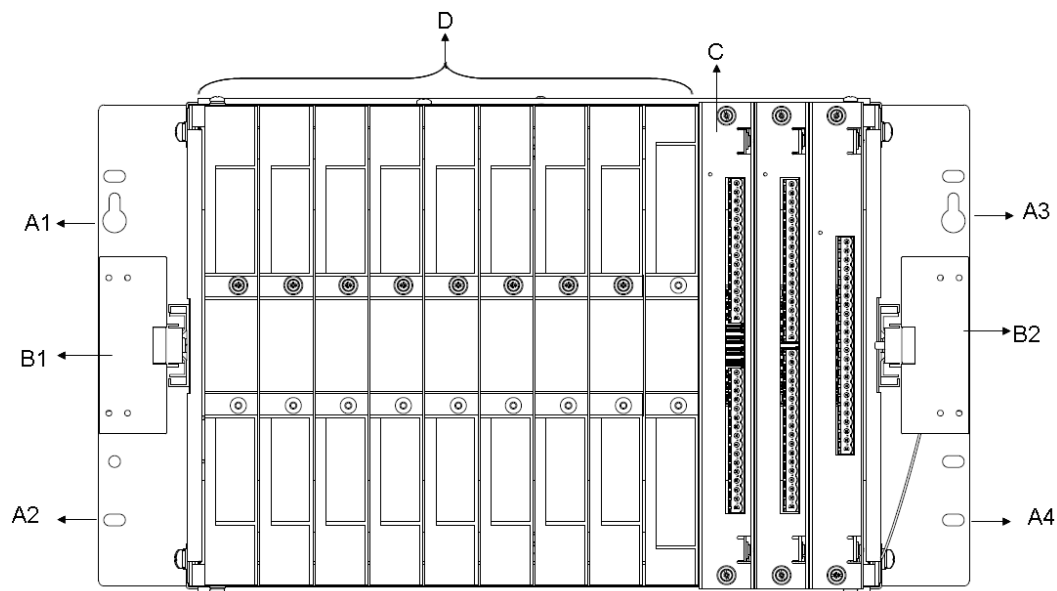
Right and Left view



1 Introduction

1.4 Wiring Kit components

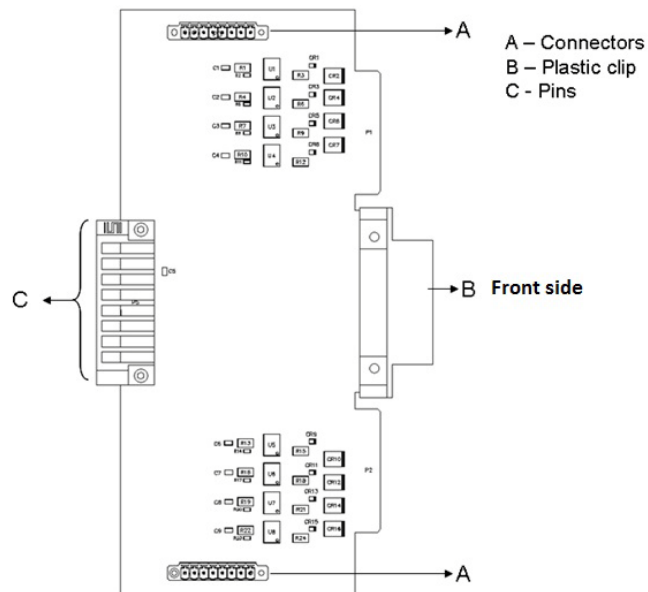
Front view



A1, A2, A3, A4 – Wiring kit mounting screws
B1, B2 – Metal clips
C – Terminal Block
D – Swing arm connectors connected to finger PCBs

Finger PCB

The finger PCBs, in the LM migration approach, connects to the Back Plane PCB at the rear of the Wiring Kit. At the front, the finger PCB connects to the Swing arm connectors/Terminal Blocks (TB) that is connected to the field devices.



There are two different types of finger PCBs – Passive and Active. You can select the PCB type based on your requirement.



ATTENTION

Choose the appropriate active board to meet the following requirements.

- Channel isolation.
- LM-MLPLC I/O card compatibility.

1 Introduction

1.4 Wiring Kit components

Passive finger PCB

The following table lists the part details for the passive finger PCB.

Description	Honeywell Part Number
6 Position Swing Arm Board	51306677-100
6 Position Swing Arm Board (for Relay Mapping)	51306677-200
11 Position Swing Arm Board	51306679-100
11 Position Swing Arm Board (for AO)	51306679-200
24 Position TB AI Board	51306681-100
16 Position TB RTD Board	51306683-100
UAIM Board	51306773-100
19 Position TB Board	51306824-100

Note:

- See the **LM_WiringKit_MLPLC_MappingList_R200.xls** file attached to this PDF for the complete Wiring Kit - modules mapping list.
- The UAIM board supports voltage and current inputs. It does not support thermocouple inputs

Active finger PCB

The following table lists the part details for the active finger PCB.

Description	Honeywell Part Number
12-24Vdc to 24Vdc 8 point DI Board	51306689-100
24Vac to 24Vdc 8 point DI Board	51306691-100
Pulse Input Module to HSC Board	51306693-100
115Vdc to 24Vdc 8 point DI Board	51306695-100
115Vdc to 24Vdc 6 point DI Board	51306697-100
115Vac/Vdc to 24Vdc 6 point Isolated DI Board	51306735-100
115Vac to 24Vdc 16 point DI Board	51306737-100

Description	Honeywell Part Number
240Vac to 24Vdc 16 point DI Board	51306737-200
12-24Vdc 8 point DO Board	51306745-100
24Vdc 16 point DO Board	51306771-100
115Vac/Vdc to 24Vdc 8 point DI Board	51306783-100
24Vdc 32 point DI Board	51306805-100
24Vdc 32 point DO Board	51306821-100

Note: See the **LM_WiringKit_MLPLC_MappingList_R200.xls** file attached to this PDF for the complete Wiring Kit - modules mapping list.

Field connectors termination

The field connectors are connected to the finger PCB of the Wiring Kit in a similar way as they are connected to the LM I/O modules. This ensures that the field wiring is not disturbed.

Swing Arm connectors are the field connectors that are hinged on the Wiring Kit. With the help of the swing arm terminal blocks, it is easy to insert, remove, and wire the modules. The edge connector of each terminal block is firmly seated onto the front edge of the circuit board and secured with the thumbscrew fastener.

The following are the different types of the field connectors that can be connected to the Wiring Kit.

- 6-position Swing Arm connector
- 11-position Swing Arm connector
- 16-position Terminal Block connector
- 19-position Terminal Block connector
- 24-position Terminal Block connector

1 Introduction

1.4 Wiring Kit components



Custom cables

Custom cables enable the Wiring Kit connections to the MLPLC I/Os. The cables are marked with the LM module names, the equivalent MLPLC module names, and the I/O module description.

The following figure illustrates a Custom cable marked with the Honeywell part number.



Unshielded custom cables

Unshielded custom cables can be used if the MLPLC I/O rack is rail mounted on the Wiring Kit or in the same panel as the Wiring Kit.



TIP

The Honeywell Part Number contains a tab value which varies for different cable lengths, as follows:

Tab Values	Measurement
100	750 mm
200	1.2 meters
300	3 meters

Substitute these tab values as per the necessary cable length.

The following table lists the part details for the unshielded custom cables.

Description	Honeywell Part Number
Cable Assembly, Unshielded, 621-1100R and 2MLI-A21A	51306709-100
Cable Assembly, Unshielded, 621-1101R and 2MLI-A21A	51306710-100
Cable Assembly, Unshielded, 621-0007R and 2MLQ-RY1A	51306711-100
Cable Assembly, Unshielded, 621-2100R/ 621-2102R and 2MLQ-SS2A	51306712-100
Cable Assembly, Unshielded, 621-2101R and 2MLQ-SS2A	51306713-100
Cable Assembly, Unshielded, 621-1160R and 2MLI-A21A	51306714-100
Cable Assembly, Unshielded, 621-3560R and 2MLI-D22A	51306715-100

1 Introduction

1.4 Wiring Kit components

Description	Honeywell Part Number
Cable Assembly, Unshielded, 621-0010AR and 2MLF-DC4A	51306716-100
Cable Assembly, Unshielded, 621-0010VR and 2MLF-DV4A	51306717-100
Cable Assembly, Unshielded, 621-2150R and 2MLQ-SS2A	51306718-100
Cable Assembly, Unshielded, 621-6550R and 2MLQ-TR2B	51306719-100
Cable Assembly, Unshielded, 621-0025R and 2MLF-RD4A	51306720-100
Cable Assembly, Unshielded, 621-0022AR and 2MLF-AC8A	51306721-100
Cable Assembly, Unshielded, 621-0022VR and 2MLF-AV8A	51306722-100
Cable Assembly, Unshielded, 621-1100R and 2MLI-D21A(DC Part)	51306723-100
Cable Assembly, Unshielded, 621-1101R and 2MLI-D21A(DC Part)	51306724-100
Cable Assembly, Unshielded, 621-0024R and 2MLF-H02A	51306725-100
Cable Assembly, Unshielded, 621-4500R and 2MLI-D21A	51306726-100
Cable Assembly, Unshielded, 621-1500R and 2MLI-D21A	51306727-100
Cable Assembly, Unshielded, 621-1100R and 2MLI-D21A(AC/DC Part)	51306774-100
Cable Assembly, Unshielded, 621-1101R and 2MLI-D21A(AC/DC Part)	51306775-100
Cable Assembly, Unshielded, 621-1160R and 2MLI-D22A	51306778-100
Cable Assembly, Unshielded, 621-6550R and 2MLQ-TR2B(Active)	51306780-100

Description	Honeywell Part Number
Cable Assembly, Unshielded, 621-6500R/ 621-6503R and 2MLQ-TR2B	51306781-100
Cable Assembly, Unshielded, 621-0020R and 2MLF-AD16A	51306784-100
Cable Assembly, Unshielded, 621-2101R and 2MLQ-RY1A	51306806-100
Cable Assembly, Unshielded, 621-2100R/ 621-2102R and 2MLQ-RY1A	51306807-100
Cable Assembly, Unshielded, 621-2150R and 2MLQ-RY2A	51306808-100
Cable Assembly, Unshielded, 621-3580R and 2MLI-D24A for DI Active	51306826-100
Cable Assembly, Unshielded, 621-3580R and 2MLI-D24A for DI Passive	51306827-100
Cable Assembly, Unshielded, 621-6575 and 2MLQ-TR4B for DO Active	51306828-100
Cable Assembly, Unshielded, 621-1250R and 2MLI-D22A	51306830-100
Cable Assembly, Unshielded, 621-2200R and 2MLQ-RY1A	51306844-100

**TIP**

See the [LM_WiringKit_MLPLC_MappingList_R200.xls](#) attached to this PDF for the complete Wiring Kit - modules mapping list.

Retractable I/O Assembly

The movable I/O base provides ease of maintenance. It provides an easy access to the finger PCB whenever there is a need for replacement. While the Wiring Kit is in service mode, the movement of the MLPLC I/O rack can be avoided by using the locking screws.

1 Introduction

1.4 Wiring Kit components

The following table lists the part details of the full rack and half rack.

Description	Honeywell Part Number
Bracket Assembly, Full Rack Wiring Kit	51454135-100
Full Rack Back Plane Board	51306675-100
Bracket Assembly, Half Rack Wiring Kit	51454139-100
Half Rack Back Plane Board	51306699-100

DC power supply distribution

The Wiring Kit requires an external +24V DC power supply if the active finger PCB is installed in the Wiring Kit.

The following is the recommended off the shelf power supply for the Wiring kit.

- Part number: HWS600-24
Manufacturer: TDK Lambda
Specification: 24V, 31 A (Peak current)

The power on/off status can be determined by measuring the voltage at the input power terminals on the backplane.

Wiring kit can be powered from the HPM panel power supply also. However, it needs additional care to avoid any possible overloading.



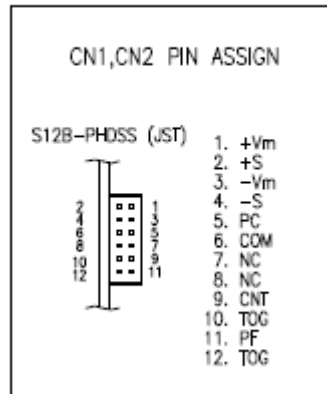
ATTENTION

- HWS600-24 power supply and the accessories needed for mounting the power supply are not part of the Wiring Kit.
 - The current consumption of each Wiring Kit depends on the I/O configuration.
 - On an average 5 full rack wiring kits or 7 half rack wiring kits can be powered by a single power supply (HWS600-24).
-

The maximum current consumption of a full rack Wiring Kit is 5A and that of a half rack Wiring Kit is 3.5A.

The following figure illustrates the signal connector information for the power supply.

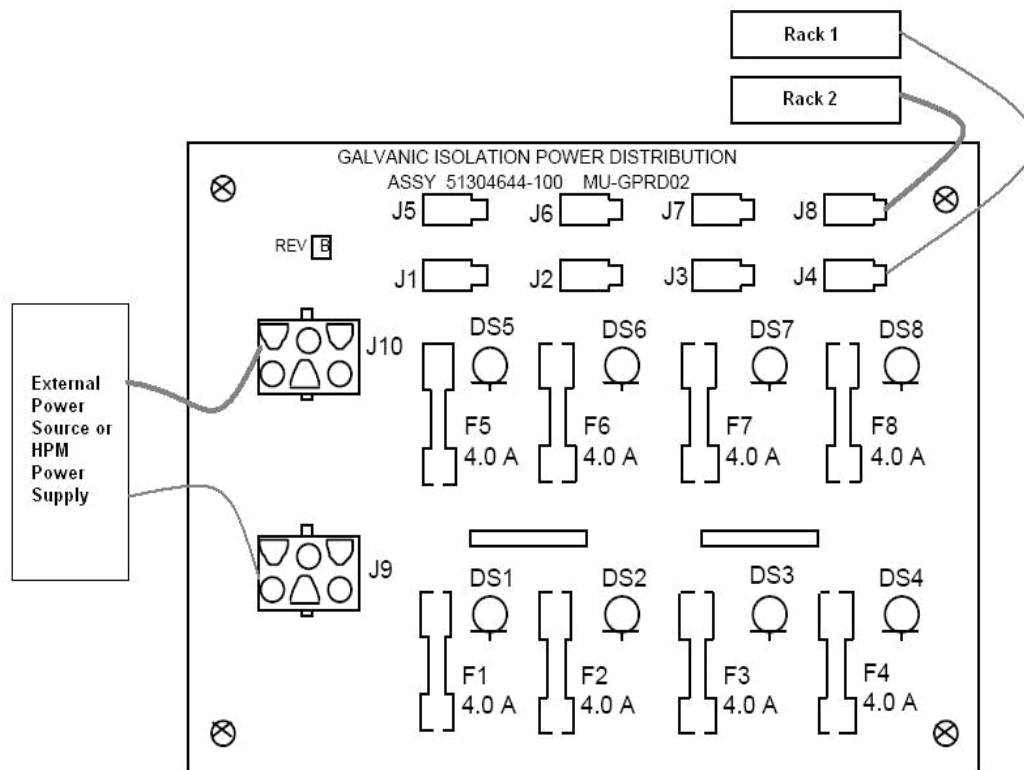
SIGNAL CONNECTOR INFORMATION



1 Introduction

1.4 Wiring Kit components

The following figure illustrates the Power Distribution Assembly and depicts the fuse rating for a HALF rack wiring kit. The TDK Lambda power supply or the HPM power supply must be connected to the power supply terminals J9/J10. Each of the J1 to J8 terminals connects to a rack to power it.



For the full rack wiring kit, the fuse rating is 4A. For the half rack wiring kit, the fuse rating is 3A.

For the distribution of power to the Wiring Kit when using an HPM power supply, the existing power supply distribution FTA is reused. For more information, see the “Power Distribution Assembly” section of the *Process Manager I/O Installation (PM/APM/HPM) Manual*.

A mounting plate is used for mounting the MLPLC processor in the rack. The additional space available in the mounting plate is used for mounting the power supply unit. For more information about mounting the power supply, see the *Mounting the power supply for the Wiring Kit* section.

The following tables list the part details for the power supply.

Description	Manufacturer Part Number
POWER SUPPLY, 24V, 27A Max, TDK LAMBDA	HWS600-24

Description	Honeywell Part Number
POWER SUPPLY ACCESSORIES	
BRACKET POWER SUPPLY ADJACENT, WIRING KIT	51454169-100
BRACKET POWER SUPPLY BOTTOM, WIRING KIT	51454170-100
POWER SUPPLY DISTRIBUTION BOARD	51304644-125
CABLE ASSY, MAINS TO POWER SUPPLY, 3MTR	51306767-100
CABLE ASSY, POWER SUPPLY TO DISTRIBUTION BOARD, 0.5MTR	51306768-100
CABLE ASSY, DISTRIBUTION BOARD TO WIRING KIT, 3MTR	51306769-100
ACCESSORIES IF HPM POWER SUPPLY IS USED	
POWER SUPPLY BRACKET	
BRACKET POWER SUPPLY ADJACENT, WIRING KIT	51454169-100
BRACKET POWER SUPPLY BOTTOM, WIRING KIT	51454170-100
POWER SUPPLY DISTRIBUTION BOARD	51304644-125
CABLE ASSY, HPM SUPPLY TO POWER DIST CARD	51306797-100
CABLE ASSY, DISTRIBUTION BOARD TO WIRING KIT, 3MTR	51306769-100



TIP

The power supply mating connector on the back plane of the wiring kit is unique to the power supply Cable assembly. The ordering Part number for the mating connector is 1826306 and manufacturer is Phoenix Contact.

1 Introduction

1.4 Wiring Kit components



TIP

This connector can be removed, to disconnect the 24Vdc from the rack.



TIP

All the wiring kit components must be stored within the storage specifications mentioned in the Specification document.

2. Planning for LM replacement

2.1 Understanding LM configuration

The following diagrams are required to understand the LM configuration.

- Panel diagram for LM.
- Rack room layout.
- Remote Terminal Panel (RTP) wiring scheme (if applicable) for LM.
- 110V wiring and 24V power supply wiring scheme.
- LM I/O kit configuration.

2.2 Hardware components

Cabinet layout

The LM processor rack is replaced by the MLPLC processor rack and the LM I/O rack is replaced by the Wiring Kit and the MLPLC I/O rack.

Compliance with standards – CE/CSA

Wiring kit must be housed in a cabinet which has a door with a lock. The normal operation of the wiring kit does not require any monitoring. Only authorized maintenance personnel must be allowed to access the cabinet for service needs. Since it is a migration wiring kit, CSA certification for the wiring kit is valid only when it is housed inside a CSA certified cabinet. In addition, the cabinet must have a safety earth connection complying with the CE/CSA standards, as applicable.



REFERENCE - INTERNAL

For more information about Cabinets, see the *Cabinet Installation* and *Card File Installation* sections in the *Logic Manager Installation*.

2 Planning for LM replacement

2.2 Hardware components



ATTENTION

The Wiring Kit requires an external +24V DC power supply if the active finger PCB is installed in the Wiring Kit.

Cable routing schemes need to be considered for custom cables with different panel and RTP. Collect the revised Panel diagram and the Wiring Kit configuration details for migration.

Finger PCBs and Custom cables

Identify the finger PCBs and the custom cables required for the replacement, based on the LM I/O type.

LM IO MODULE	CUSTOM CABLE	FINGER PCB	MLPLC IO MODULE
621-1100R	51306709-100	51306677-100	2MLI-A21A
621-1100R	51306723-100	51306695-100	2MLI-D21A
621-1100R	51306774-100	51306783-100	2MLI-D21A
621-1101R	51306710-100	51306677-100	2MLI-A21A
621-1101R	51306724-100	51306697-100	2MLI-D21A
621-1101R	51306775-100	51306735-100	2MLI-D21A
621-1160R	51306714-100	51306679-100	2MLI-A21A
621-1160R	51306778-100	51306737-100	2MLI-D22A
621-1500R	51306727-100	51306691-100	2MLI-D21A
621-3552R	51306715-100	51306679-100	2MLI-D22A
621-3560R	51306715-100	51306679-100	2MLI-D22A
621-4500R	51306726-100	51306689-100	2MLI-D21A
621-0007R	51306711-100	51306677-200	2MLQ-RY1A
621-2100R	51306712-100	51306677-100	2MLQ-SS2A
621-2100R	51306807-100	51306677-200	2MLQ-RY1A
621-2101R	51306713-100	51306677-100	2MLQ-SS2A
621-2101R	51306806-100	51306677-200	2MLQ-RY1A
621-2150R	51306718-100	51306679-100	2MLQ-SS2A
621-2150R	51306808-100	51306679-100	2MLQ- RY2A
621-6550R	51306719-100	51306679-100	2MLQ-TR2B
621-6550R	51306780-100	51306771-100	2MLQ-TR2B
621-6500R	51306781-100	51306745-100	2MLQ-TR2B

2 Planning for LM replacement

2.2 Hardware components

LM IO MODULE	CUSTOM CABLE	FINGER PCB	MLPLC IO MODULE
621-6503R	51306781-100	51306745-100	2MLQ-TR2B
621-0022-VR	51306722-100	51306681-100	2MLF-AV8A
621-0010-AR	51306716-100	51306679-200	2MLF-DC4A
621-0010-VR	51306717-100	51306679-200	2MLF-DV4A
621-0024R	51306725-100	51306693-100	2MLF-H02A
621-0025R	51306720-100	51306683-100	2MLF-RD4A
621-0020R	51306784-100	51306773-100	2MLF-AD16A
621-2102R	51306712-100	51306677-100	2MLQ-SS2A
621-2102R	51306776-100	51306741-100	2MLQ-TR2B
621-2102R	51306807-100	51306677-200	2MLQ-RY1A
621-2200R	51306844-100	51306677-200	2MLQ-RY1A
621-1250R	51306830-100	51306737-200	2MLI-D22A
621-3580	51306827-100	51306824-100	2MLI-D24A
621-3580	51306826-100	51306805-100	2MLI-D24A
621-6575	51306828-100	51306821-100	2MLQ-TR4B

Note: See the **LM_WiringKit_MLPLC_MappingList_R200.xls** file attached to this PDF for the complete Wiring Kit - modules mapping list.

LM and ML PLC I/O map

Before migrating the LM I/Os, identify the equivalent ML PLC I/Os.

For identifying the equivalent ML PLC I/O modules for each of the LM I/O module see the **LM_WiringKit_MLPLC_MappingList_R200.xls** file attached to this PDF.

Every LM rack consists of a parallel/serial communication module, a power supply module and an enhanced diagnostic module. The functions of these modules are supported in different ways in the migration. Hence, these modules need not be considered for migration and these slots remain empty.

Regrouping of LM I/O channels

The number of I/O channels supported by the LM (to be replaced) and the corresponding MLPLC (replacement for LM) might not be the same. In such scenarios, the LM I/O channels must be regrouped in the MLPLC I/O rack.

Scenario	Solution
Number of channels supported in the LM module > Number of channels supported in the MLPLC module.	The I/O channels from an LM I/O module is connected to the equivalent identified two adjacent MLPLC I/O modules. The MLPLC I/O modules can reside in the same I/O rack or in the adjacent racks.

2 Planning for LM replacement

2.2 Hardware components

3. Replacement Tasks

3.1 Pre-replacement checklist

The existing configuration and the vital information residing on LM must be preserved before shutdown, and suitably reapplied after the ML PLC installation. The following checklist assists you in this task.

Step	Task	Done?
1	Understand all the standard conventions of your plant and collect the relevant documents. So that, the newly added information is in synchronization with the old data. Collect the standard conventions for the following: - Ferrule nomenclature - Cable tags nomenclature - Tag naming conventions for I/O	
 REFERENCE - INTERNAL For more information about wiring, see the <i>Wiring</i> section in the <i>Logic Manager Installation</i>.		
2	Collect the control system specification document (write-up) of the plant to know the existing control configuration of the plant.	
3	Collect the relevant drawings for the existing LM configuration.	
4	Collect the existing I/O database details of the plant.	
5	Collect the peer connection list and peer to peer design concepts of the LM.	
6	Collect the list of all displays/graphics containing the LM points. If the box parameters are directly used for the display, then those displays need updates.	
7	Verify the existing TPN release number.	
8	Extract the EB files of the LM identified for replacement. The EB files are required to obtain the point details, I/O module details, and node specific configuration details of the LM.	

3 Replacement Tasks

3.1 Pre-replacement checklist

Step	Task	Done?
9	Identify the mode of communication: RS232 or RS485.	
10	Checkpoint the LM nodes identified to be involved in the migration process.	
	REFERENCE - INTERNAL <i>For more information about checkpointing, see the Process Operations Manual SW11-601.</i>	
11	Extract the LM Printable Ladder (.prn) files for logics.	
12	From the EB files, extract the I/O details and the node specific configuration details of the LM.	
13	Convert the existing I/O and the LM points into suitable MLPLC points, ladder configuration.	
14	Identify the LM modules that are excluded in the migration and leave them untouched, until otherwise alternates are stated in Honeywell references or guided by Honeywell personnel.	
15	Ensure that the LM is ready for an offline migration.	
16	Completely delete the LM database from the system to avoid existence of ghost points.	
17	Shutdown the LM and proceed with the migration.	

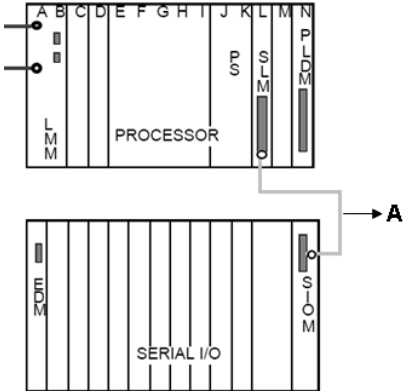
3.2 Replacing the LM processor

Prerequisite

The plant must be shutdown before performing the migration procedure. As the Wiring Kit retains the field wiring intact, the shutdown time for this migration is minimal.

Replacing the LM processor with the MLPLC processor

Perform the following steps to remove the LM processor and replace the MLPLC processor.

Step	Action
	WARNING Ensure that the identified LM, which is to be replaced, is shutdown, turned off and disconnected from the power supply.
Remove the LM Processor Rack	
1	Remove the serial communication connection between the LM I/O module and the LM processor module.
	
A – I/O wiring communication between the LM processor and the LM I/O module	

3 Replacement Tasks

3.2 Replacing the LM processor

Step	Action
	TIP • Preserve the Serial communication module with the rack installation since this module has all the address settings applicable to that particular rack. • The switches will be configured for particular applications and needs to be preserved.
	REFERENCE - INTERNAL For more information, refer to the <i>Serial Link Module section of Logic Manager Service Manual</i>, LM13-500.
2	Remove UCN Drop Cables.
3	If the LM is redundant, remove the LM Redundancy wiring from the LMM and RCM modules, and the redundancy cable from the RCM module.
	REFERENCE - INTERNAL For more information, refer to the <i>Universal Control Network Planning</i>, UN02-501.
4	Unscrew the mounting screws and remove the LM processor.
Mount the MLPLC Processor Rack	
1	On the panel, mount the MLPLC mounting plate in place of the removed LM processor.
	TIP The MLPLC mounting plate is applicable only for mounting the MLPLC processor in an existing panel.
2	Mount the MLPLC processor on the MLPLC mounting plate. Note: If the ML is redundant, repeat steps 1 and 2 to mount the redundant ML.
3	Connect the power supply to the MLPLC processor.

3.3 Replacing the LM I/O rack

Removing the local LM I/O rack

Perform the following steps to remove the local LM I/O rack.

Step	Action
	ATTENTION Disconnect the power cord to the LM I/O rack Power Supply module from the power source. Ensure that the communication wiring between the LM processor and the LM I/O module is removed.
1	Identify the field cables to be removed.
CAUTION	Verify the field cable tags and ferrules to ensure that they match with the wiring diagrams.
2	Remove the LM I/O serial communication module from the LM I/O rack.
3	Mark the I/O rack details (from where the module is removed) on the module for reference.
4	Remove the swing arm connectors or terminal block connectors of the field cables sequentially from all the LM modules. The swing arm connectors hang in front of the LM I/O rack.
5	Safely unscrew the mounting screws and remove the LM I/O rack.

Note: LM Wiring kit does not require any special handling care.

3 Replacement Tasks

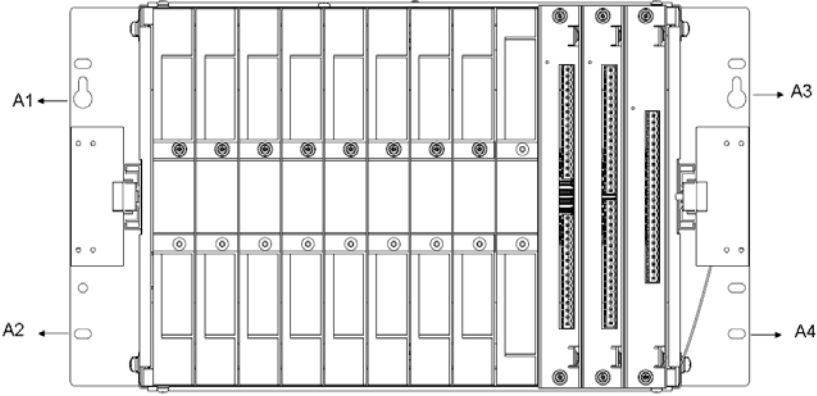
3.3 Replacing the LM I/O rack

Mount the MLPLC I/O Rack

Perform the following steps to mount the MLPLC I/O rack.

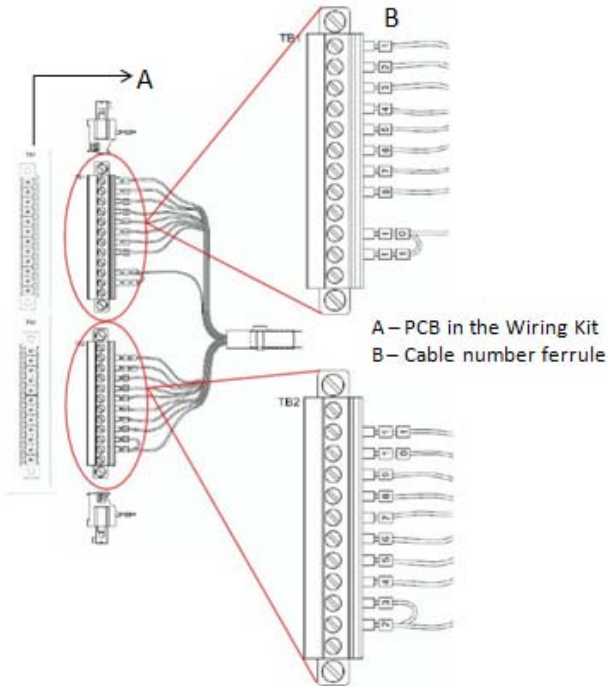
Note: The illustrations used in this procedure refer to the full rack Wiring Kit. Similarly, the half rack Wiring Kit can also be used for migration.

Step	Action
1	Assemble the back plane on the Wiring Kit using the screws in the bag Assembly, P/N 51202597-100.
2	Mount the Wiring Kit in the place of the LM I/O rack using the mounting screws.

A technical diagram of the MLPLC I/O rack assembly. It shows a rectangular frame with a grid of slots. On the left side, there are two screws labeled A1 and A2. On the right side, there are two screws labeled A3 and A4. The diagram illustrates the mounting of the wiring kit onto the back plane of the rack.

A1, A2, A3, A4 – Wiring kit mounting screws

3	Refer to the cable tags and the ferrules to identify the finger PCBs to be inserted in the Wiring Kit.
---	--

Step	Action
	 A – PCB in the Wiring Kit B – Cable number ferrule
	TIP If the cable tags are not available, identify the finger PCBs with the help of the I/O card details.
	Note • Assess the cabinet space for deciding the mounting option of MLPLC IO. • Assess the cabinet space for deciding the bottom or front entry of the custom cables.
4	Identify the custom cables to be connected to the finger PCBs.

3 Replacement Tasks

3.3 Replacing the LM I/O rack

Step

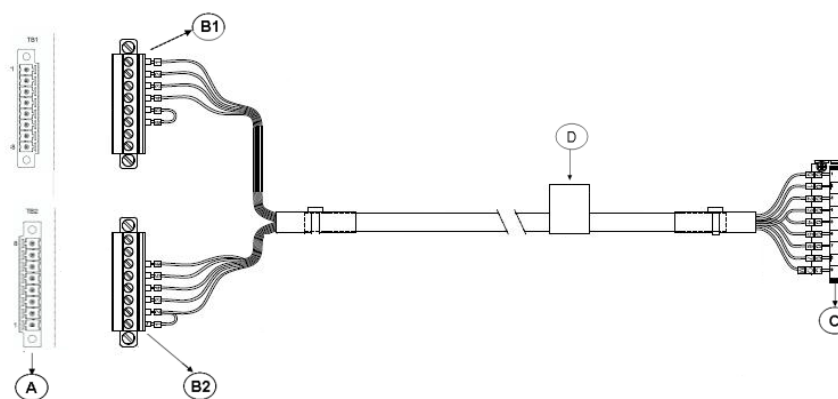
Action



TIP

The Finger PCB and Custom cable mapping helps in identifying the following:

- The finger PCBs to be inserted in the Wiring Kit.
- The custom cables to be connected to the finger PCBs.



A - Wiring Kit PCBA
B1, B2 - Connects to finger PCBs of the Wiring kit
C - Connects to ML PLC I/O Slot
D - Part number information

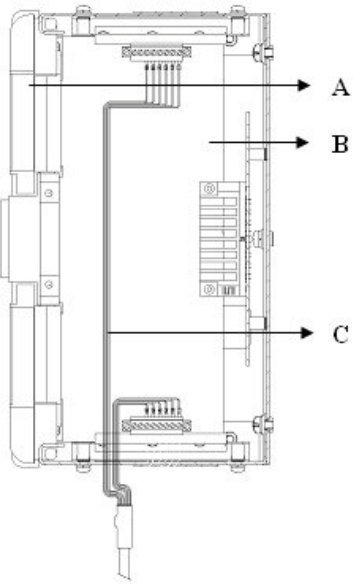
One end of the custom cable (B1, B2) is connected to the finger PCB. The other end of the custom cable (C) is connected to the MLPLC I/O rack.

5

Insert the custom cable through the bottom of the Wiring Kit.

3 Replacement Tasks

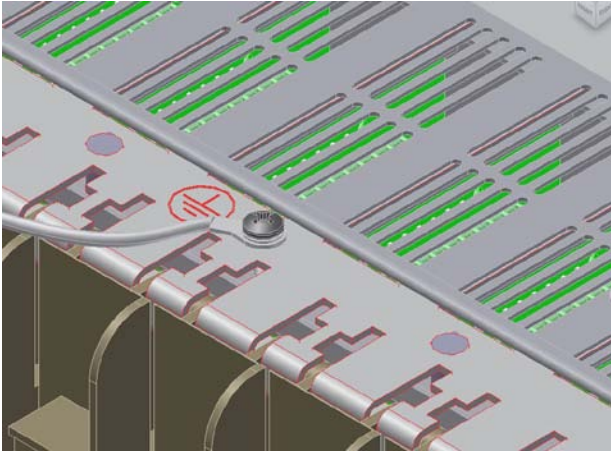
3.3 Replacing the LM I/O rack

Step	Action
	 A – Wiring Kit B – PCB C – Custom Cable
6	Connect the lengthier wire of the Connectors to the upper end of the finger PCB and the shorter wire of the Connectors to the lower end of the finger PCB.
	WARNING Avoid direct contact with the exposed pins of the finger PCBs or the Terminal Blocks, while connecting or removing the custom cables.
7	Once the cables are connected, close the bottom cover with the ESD cover, P/N 51306728-100 for Full Rack and P/N 51306729-100 for Half Rack.

3 Replacement Tasks

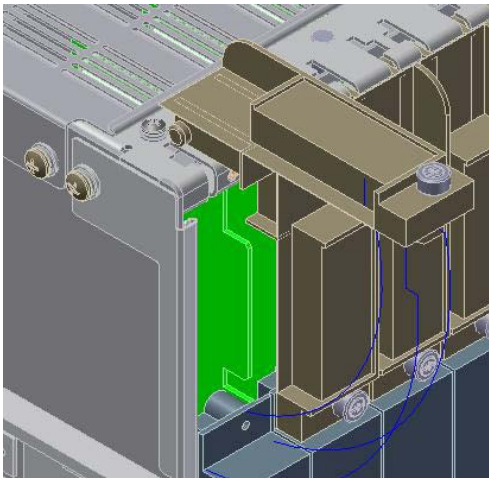
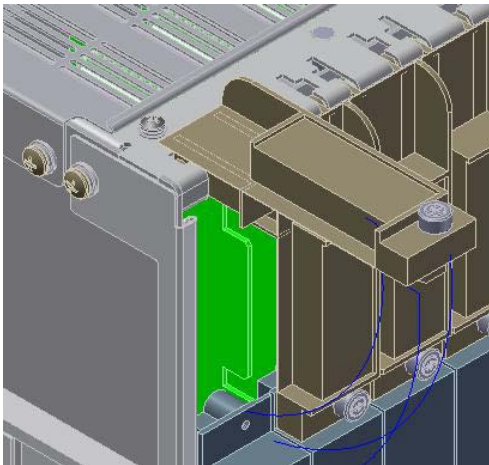
3.3 Replacing the LM I/O rack

Step	Action
	A - Wiring Kit PCB B - Upper end of the PCB C - Lower end of the PCB D - Custom cable E - Connector of the Custom cable (long wire) F - Connector of the Custom cable (short wire)
8	Insert the finger PCBs in the corresponding slot of the Wiring Kit.
CAUTION	Ensure to position the finger PCBs in the right direction before inserting them in the slot. There are markings on the PCBs indicating the top and the bottom.
	ESD HAZARD Fasten the Electro Static Discharge (ESD) cover to the chassis on the top and bottom.

Step	Action
	Note: Wiring kit is to be connected to the safety earthing system of the Plant, using a cable as recommended in the statutory standards of the region. Provision for such a connection is available on top of the kit as shown in the following Figure. 
9	Connect the swing arm connectors (hanging from the field wiring) to the corresponding finger PCBs on the Wiring Kit. Note: It is preferable to fix the top swing arm connector first.

3 Replacement Tasks

3.3 Replacing the LM I/O rack

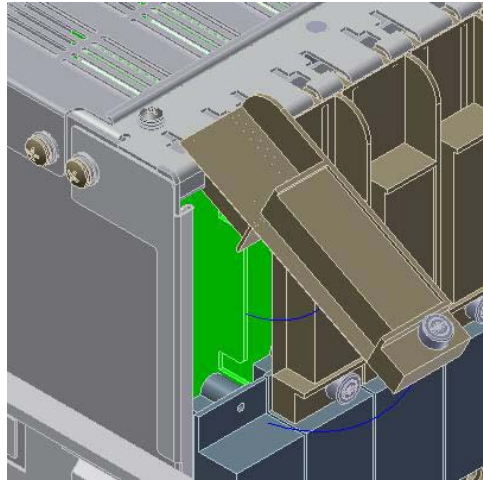
Step	Action
Step 1	
Step 2	

3 Replacement Tasks

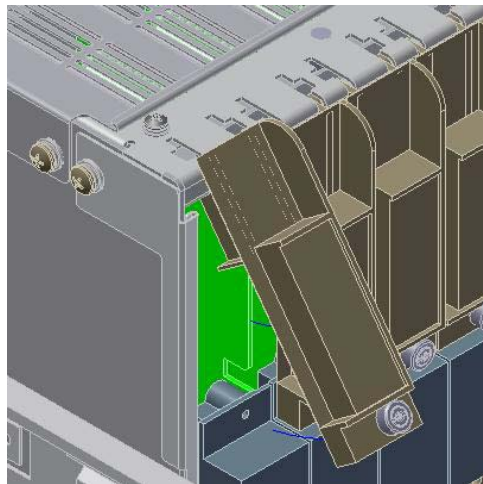
3.3 Replacing the LM I/O rack

Step	Action
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Step 3	
--------	--



Step 4	
--------	--



3 Replacement Tasks

3.3 Replacing the LM I/O rack

Step	Action
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Step 5	
--------	--



ATTENTION

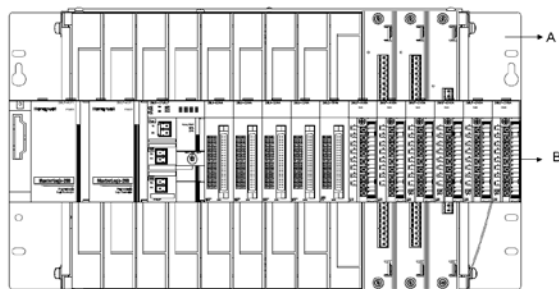
Ensure the cables or the wires connected between the Wiring Kit and the MLPLC I/O modules are not tangled.

10	Mount the MLPLC I/O rack in front of the Wiring Kit.
----	--



TIP

Alternately the MLPLC IO rack can be mounted adjacent to the panel.

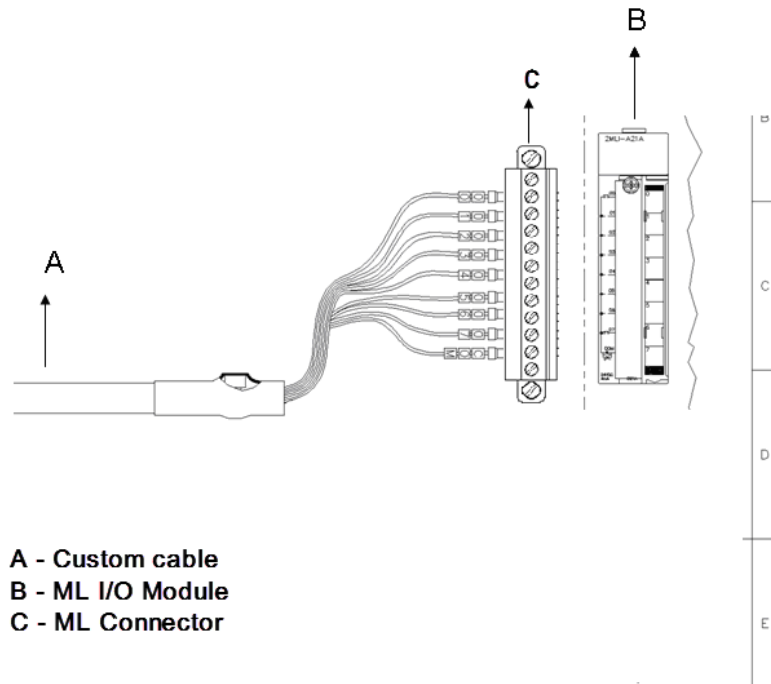


A – Wiring kit
B – ML I/O rack

3 Replacement Tasks

3.3 Replacing the LM I/O rack

Step	Action
11	Connect the MLPLC Connectors of the custom cables to the corresponding I/O modules of the mounted MLPLC I/O rack.

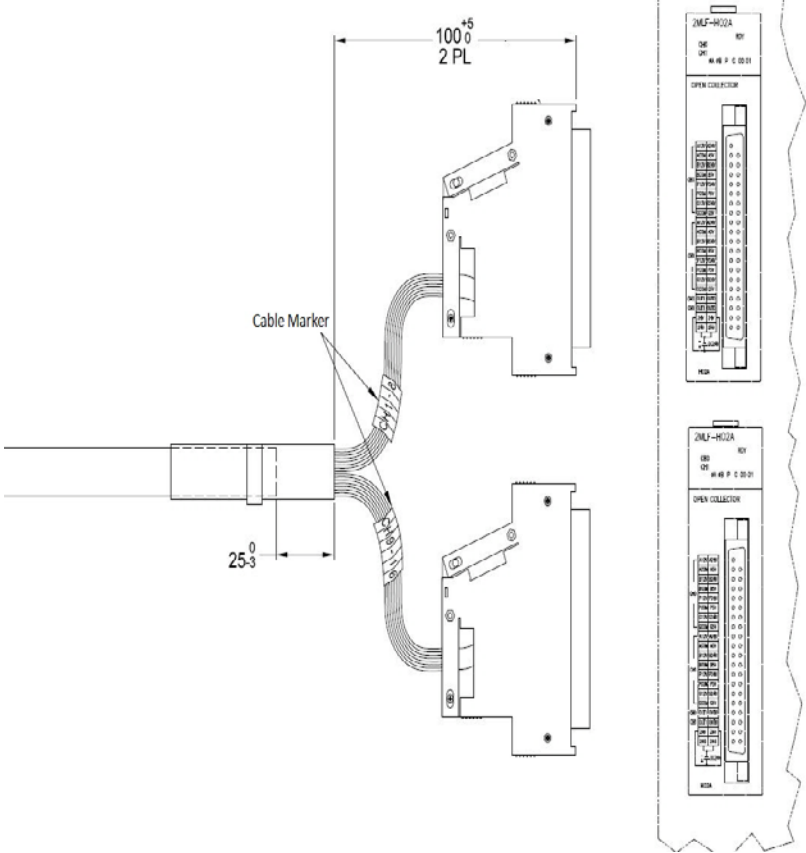


The following module combinations will terminate in two ML modules. Identify them using the cable marker. Channel 1-8 is to be connected to the left and channel 9-16 is to be connected to the right.

LM IO Module	Custom Cable	Finger PCB	MLPLC IO Module
621-1160R	51306714-100	51306679-100	2MLI-A21A
621-0025R	51306720-100	51306683-100	2MLF-RD4A
621-0024R	51306725-100	51306693-100	2MLF-H02A

3 Replacement Tasks

3.3 Replacing the LM I/O rack

Step	Action
	



ATTENTION

You must connect the MLPLC connectors to the adjacent MLPLC I/O module, only after mounting the second MLPLC I/O module.



REFERENCE - INTERNAL

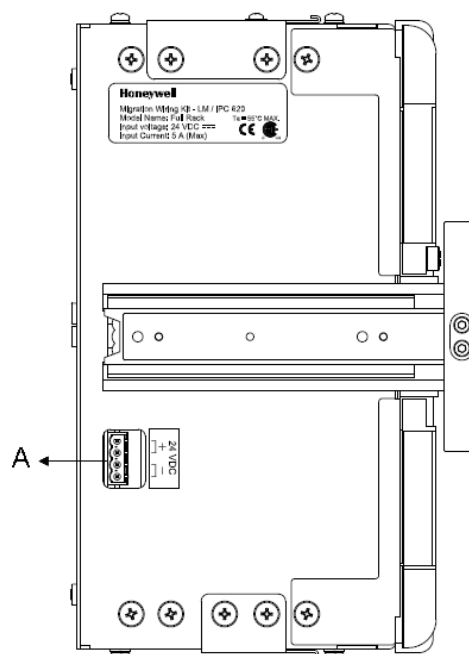
For more information on the existing connections to MLPLC, if any, refer to *Pre-replacement checklist*.

Step	Action
	TIP • If there are wires/cables already connected to the MLPLC, remove the wirings and connect the custom cables. • The number of points supported by the LM module and the MLPLC module may not be same. In such scenarios, the LM I/O channels must be regrouped. See Regrouping the LM I/O channels.
12	Repeat steps 1 through 9 for all the MLPLC I/O modules to complete the migration.
13	Connect the power supply for the Wiring Kit as per the polarity indication.
	ATTENTION Ensure that the Wiring Kit is powered by HWS600-24 or HPM power supply. The cards will not function as intended in the following cases. • Polarity reversal • Improper voltage levels beyond the tolerance limits • Improper lead connection

3 Replacement Tasks

3.3 Replacing the LM I/O rack

Step	Action
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A – Power connector on the side wall of the wiring kit



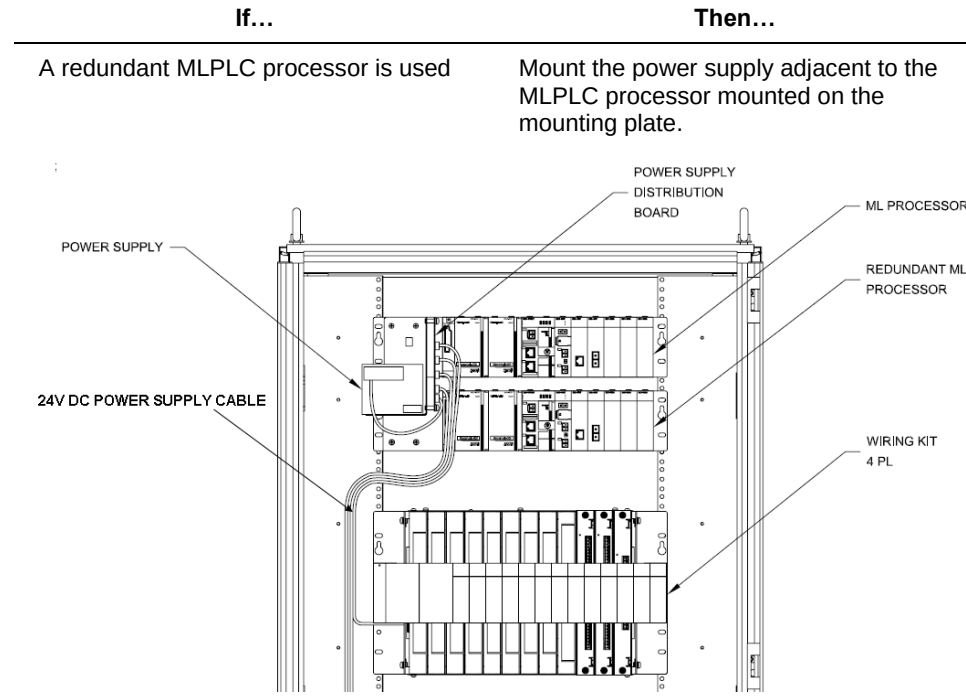
REFERENCE - INTERNAL

Proceed with the post-migration checks. See **Note: Identify the additional space required for regrouping, considering the length of the custom cable.**

Post-migration checks section.

Mounting the power supply for the Wiring Kit

If an external power supply is required for the Wiring Kit, it is mounted as illustrated.



3 Replacement Tasks

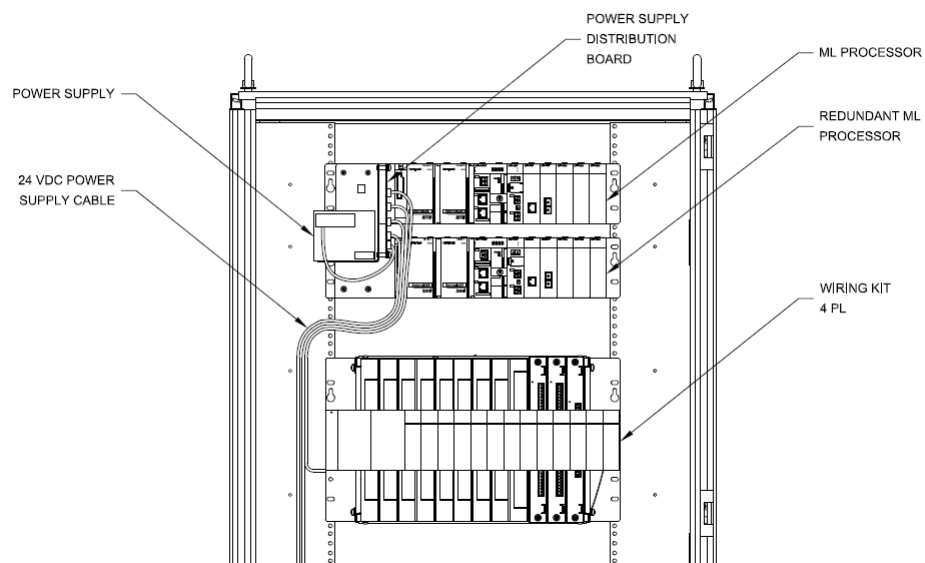
3.3 Replacing the LM I/O rack

If...

A non-redundant ML PLC processor is used

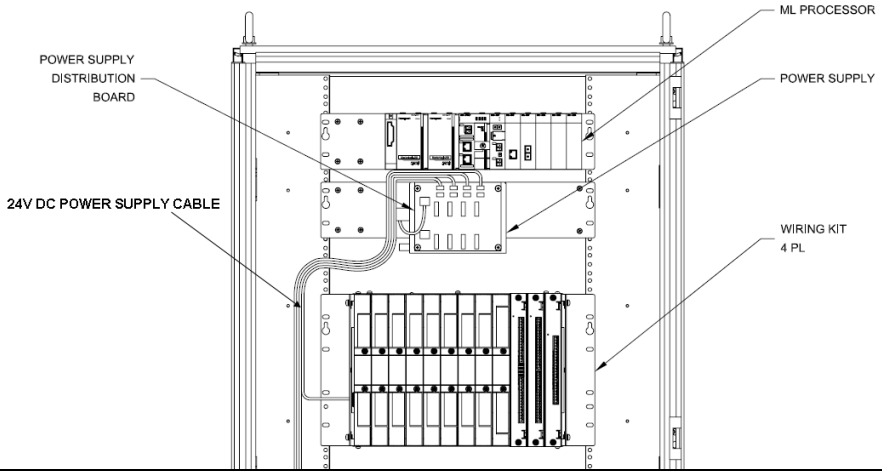
Then...

Mount the power supply adjacent to the MLPLC processor mounted on the mounting plate.



3 Replacement Tasks

3.3 Replacing the LM I/O rack

If...	Then...
• A non-redundant MLPLC processor is used. • There is a constraint in the depth of the cabinet.	Mount the power supply below the MLPLC processor mounted on the mounting plate.
 The diagram shows a side view of an MLPLC rack. At the top, an 'ML PROCESSOR' is mounted. Below it, a 'POWER SUPPLY' is mounted. To the left, a 'POWER SUPPLY DISTRIBUTION BOARD' is connected to a '24V DC POWER SUPPLY CABLE'. At the bottom, a 'WIRING KIT 4 PL' is shown. The rack is filled with various modules, including I/O modules and a power supply module.	
MLPLC I/O modules are mounted remotely.	Mount the power supply on the HPM panel, nearest to the remote I/O racks.

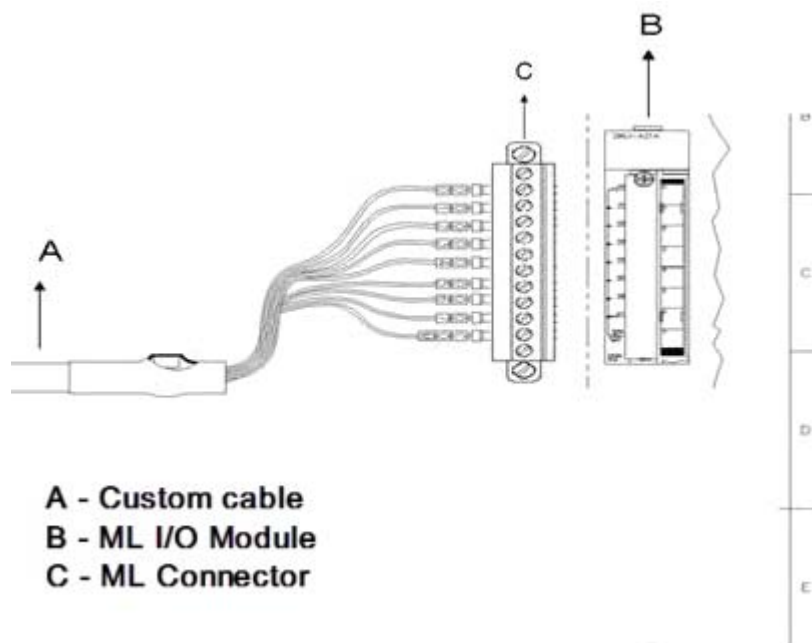
3 Replacement Tasks

3.3 Replacing the LM I/O rack

Regrouping the LM I/O channels

Perform the following steps to regroup the LM I/O channels.

Step	Action
1	Identify the custom cables that must be used for regrouping the LM I/O channels.
2	Connect one end of the custom cable to the finger PCBs in the Wiring Kit.
3	Connect the other end of the custom cables to the adjacent identified MLPLC I/O modules.



Note: Identify the additional space required for regrouping, considering the length of the custom cable.

3.4 Post-migration checks

Perform the following steps to verify the LM migration.

Step	Task	Done?
1	Ensure all the LM processors are replaced with the respective MLPLC processors.	
2	Ensure all the LM I/Os are connected to the corresponding finger PCBs on the Wiring Kit.	
3	Ensure the entire finger PCBs on the Wiring Kit are connected to the corresponding MLPLC I/O modules.	
4	Ensure all the cabling is intact and there are no freely hanging cables, unless intended to be so.	
5	Connect all the power supplies to the power sources.	
6	Perform the basic power supply tests.	
7	Perform the Loop checks.	
8	Perform the software migration check for the correctness of the data migration.	
9	Check for the changed displays and their correctness.	
10	Rebuild all the points in other nodes (xPMs/AM) accessing the LM points.	
11	Check for the redundancy requirement of the migrated modules.	

Honeywell recommends that any I/O to be added in the future can be connected to the MLPLC directly, instead of adding a PCB in the wiring kit.

3 Replacement Tasks

3.4 Post-migration checks

4. Service and maintenance

4.1 Maintaining the MLPLC processor



REFERENCE - INTERNAL

For more information about maintaining the MLPLC processor, see the *Programmable Logic Controller MasterLogic-200R User's Guide*.

4.2 Maintaining the Wiring Kit



ESD HAZARD

This equipment is subject to damage by Electrostatic Discharge (ESD). Wear your ESD wrist strap before you make contact with the equipment.



WARNING

Authorized personnel alone should take the wiring kit to service mode and access wiring kit, custom and IO cables and ML.

Servicing the Wiring Kit

The following are the two different working modes of the Wiring Kit.

- Active – The Wiring Kit is turned on and running.
- Service – The Wiring Kit is under maintenance.

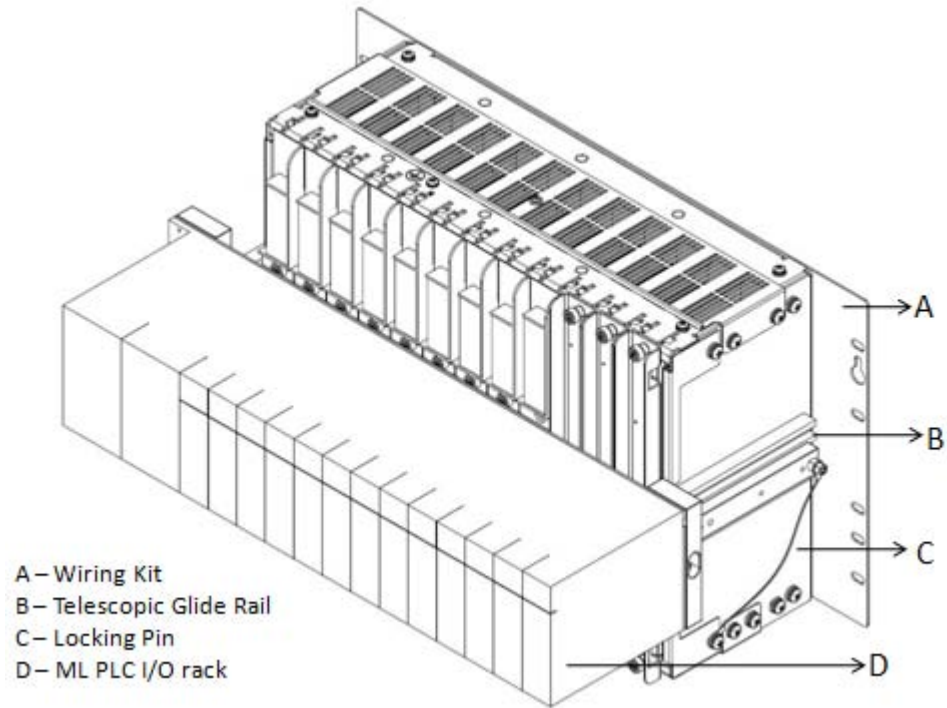
The sliding Telescopic Glide Rails are used for changing the working mode of the Wiring Kit.

4 Service and maintenance

4.2 Maintaining the Wiring Kit

Wiring Kit in active mode

The following figure illustrates the normal operating mode of the Wiring Kit.

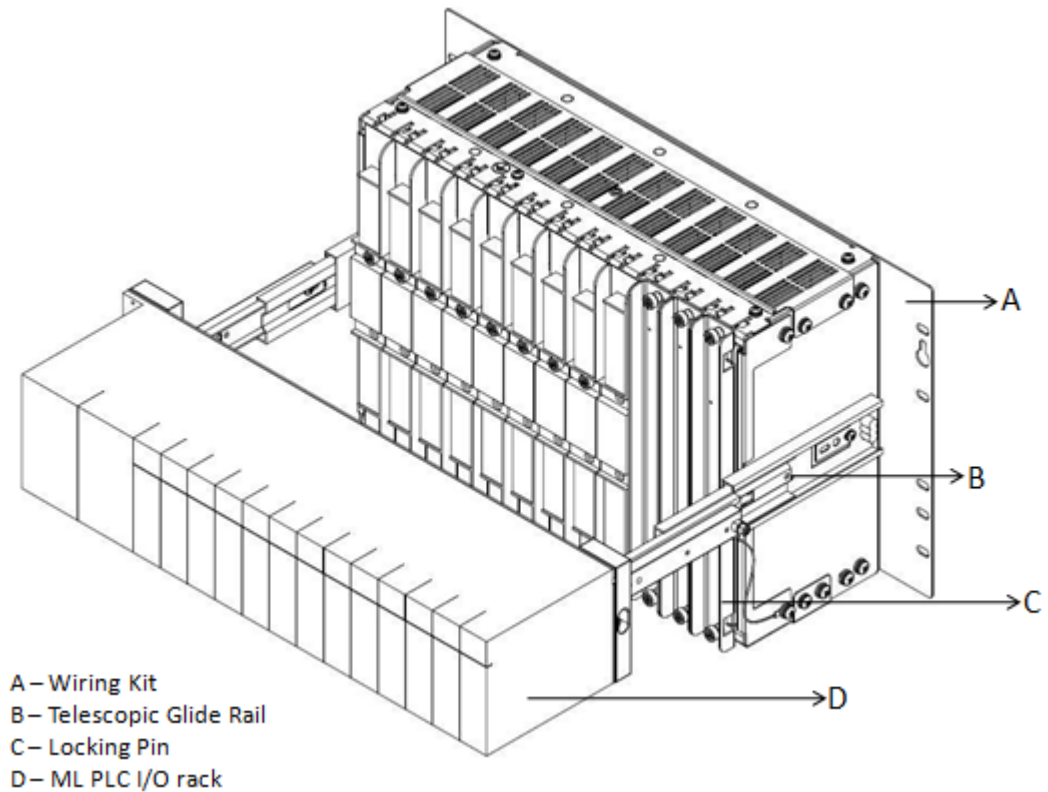


Wiring Kit in service mode

Perform the following steps to switch the Wiring Kit from active to service mode.

- If an external power supply is connected to the power connector on the left side of the Wiring Kit rack, it should be disconnected.
- Unlock the locking screws
- Slide the MLPLC I/O rack outwards on the Telescopic Glide Rails

The following figure illustrates the maintenance mode of the Wiring Kit.



Diagnosing the fault

To recover the system from the fault immediately, it is most important to quickly detect the potential causes of a fault and take corrective measures. To troubleshoot the system correctly, check the following manually.

- Operation status (Stop and Run).
- Power On/Off status.
- I/O device status.
- Wiring status (I/O wiring, extension, and communication cable).

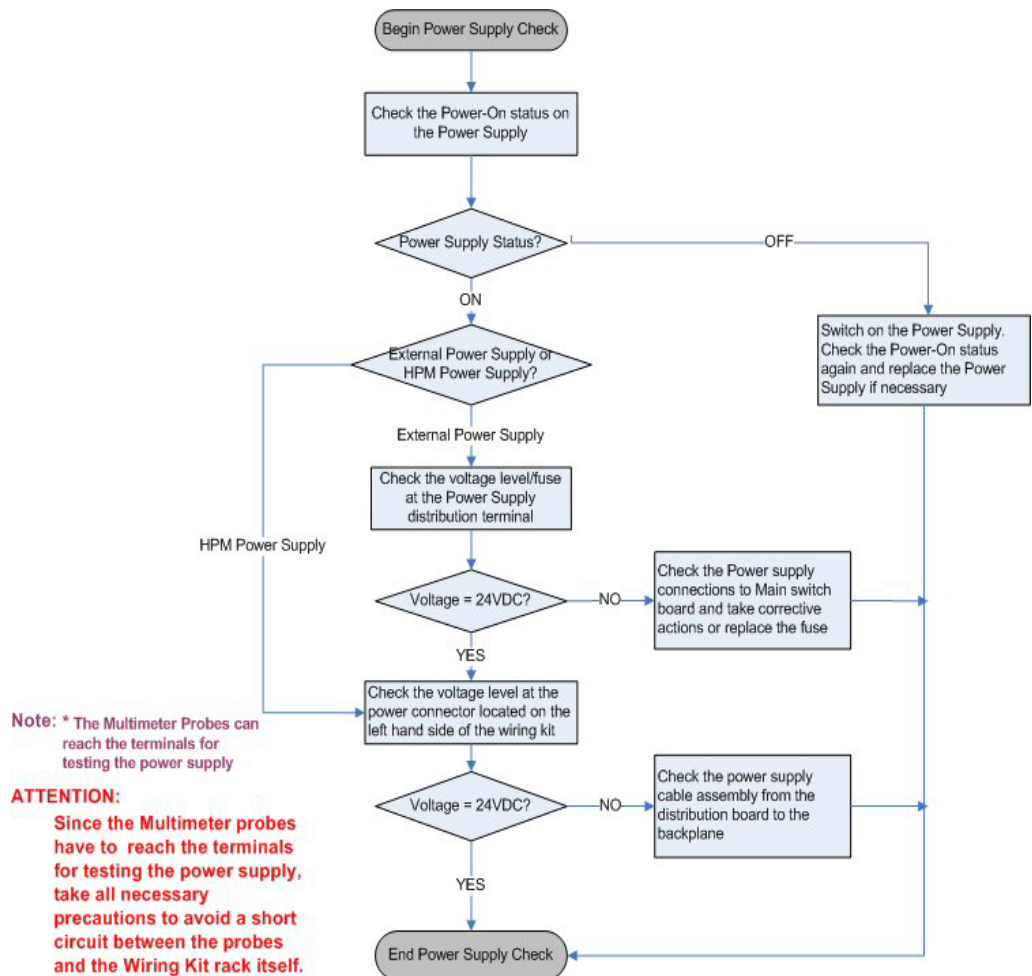
Any skeptical behavior of the IO connected is to be diagnosed using the indications of the MLPLC IO module and with the help of the SOFTMASTER.

Key points to remember while diagnosing the potential cause for the fault

- Remember passive finger PCBs are as good as wires. Simulation of the inputs at the swing arm connector terminal helps in identifying whether the defect is in the Wiring Kit or MLPLC or in the field. Field wiring may be isolated as per the requirements. However this is optional for DI.
- In the case of outputs, the field wiring should be isolated. The output state can be toggled from MLPLC and can be measured at the swing arm connector terminal.
- If many of the IOs connected through the active finger PCBs are doubted to be not working, check the power supply and the connection to the Wiring Kit rack.
- If an individual channel is to be tested, then measure the voltage of the Wiring Kit at the MLPLC IO end of custom cable. This measurement reveals the state of the channel.

The flowcharts in the following pages illustrate the procedure to diagnose the fault and the steps to restore the operation.

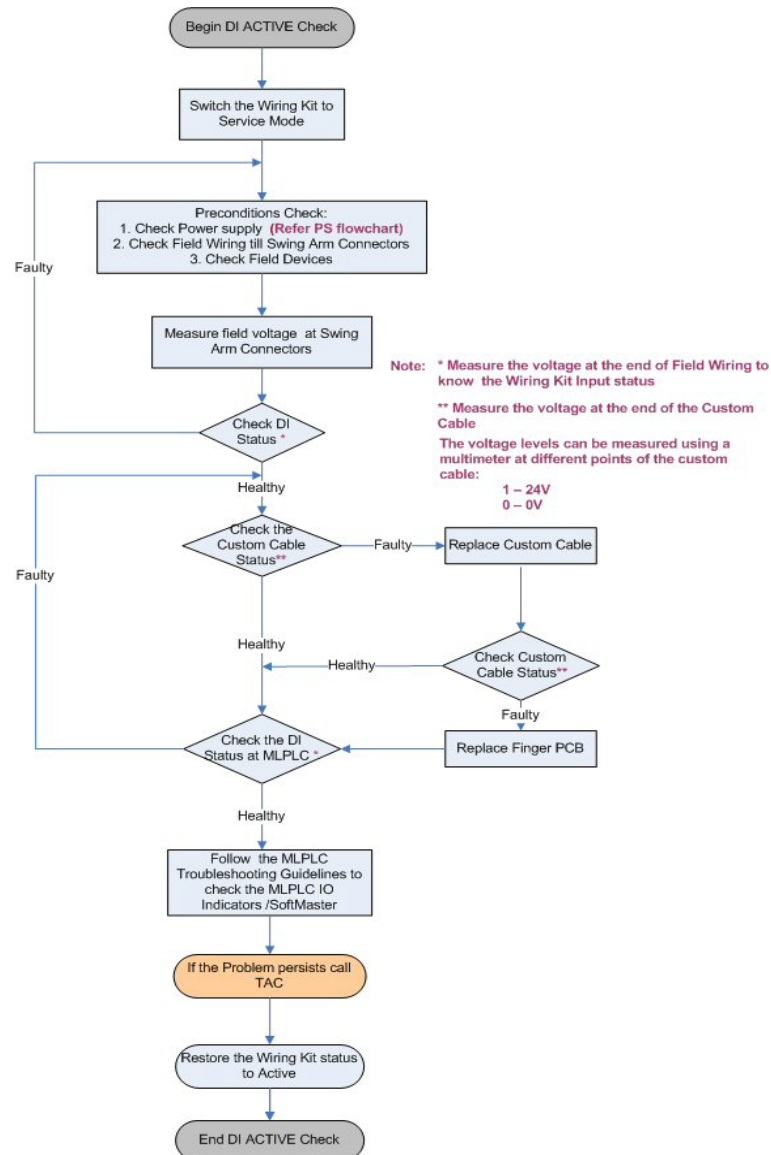
Perform the steps mentioned in the flowchart to check the power supply.



4 Service and maintenance

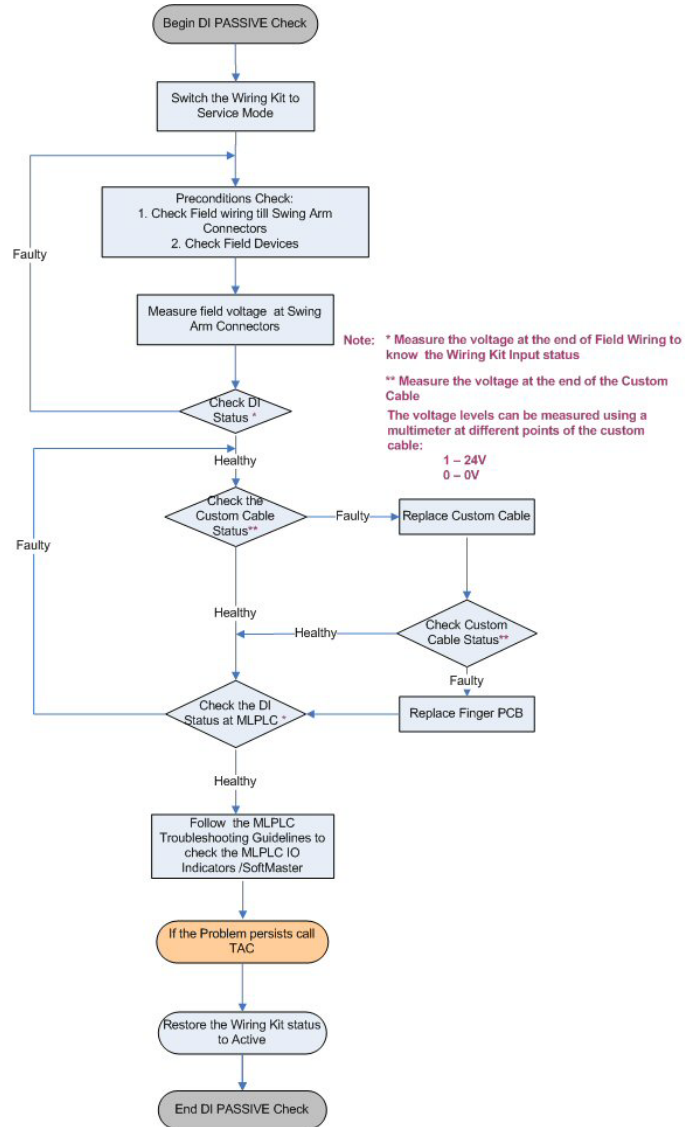
4.2 Maintaining the Wiring Kit

If a failure is detected on one or more DI channels associated with an Active Finger PCB, perform the steps mentioned in the flowchart for troubleshooting.



Note: Always terminate the ML end of the custom cable first.

If a failure is detected on one or more DI channels associated with a Passive Finger PCB, perform the steps mentioned in the flowchart for troubleshooting.

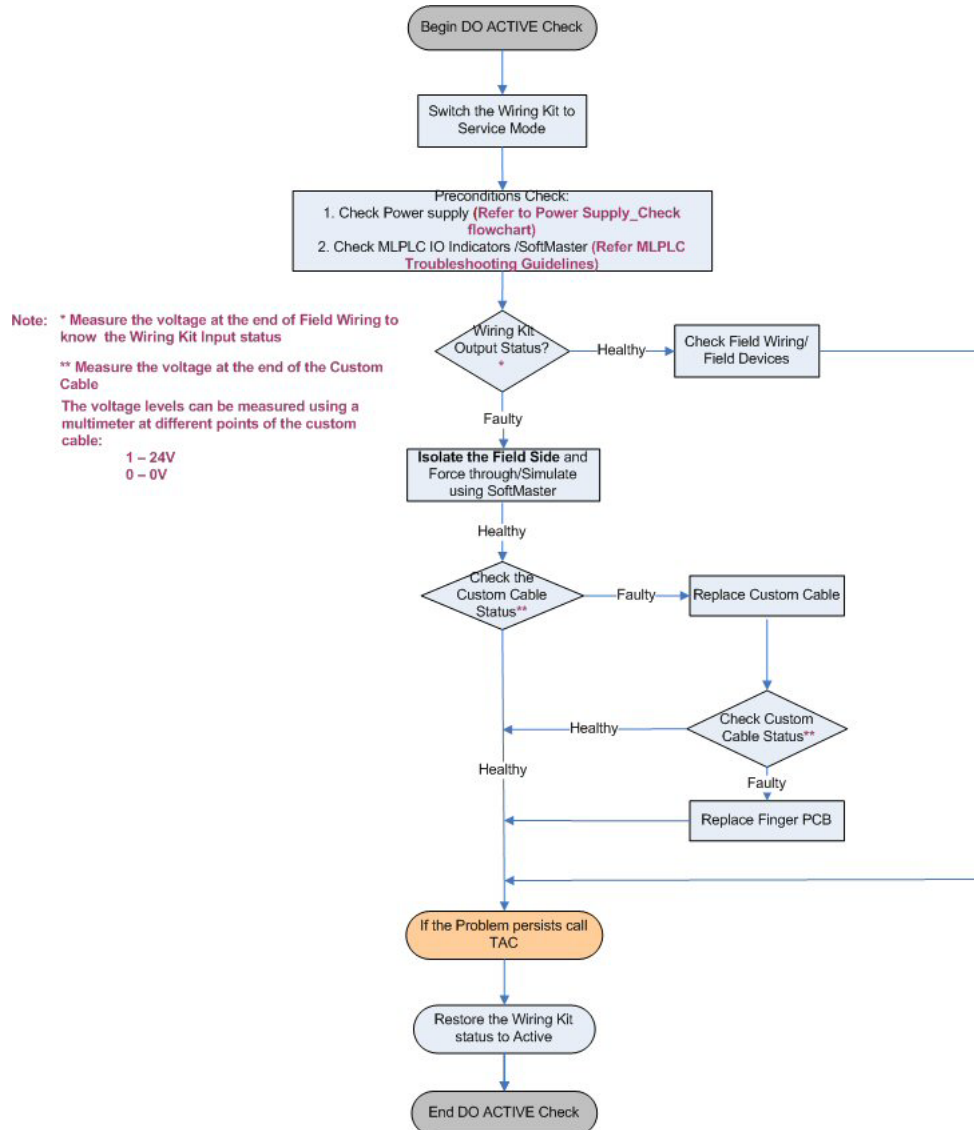


Note: Always terminate the ML end of the custom cable first.

4 Service and maintenance

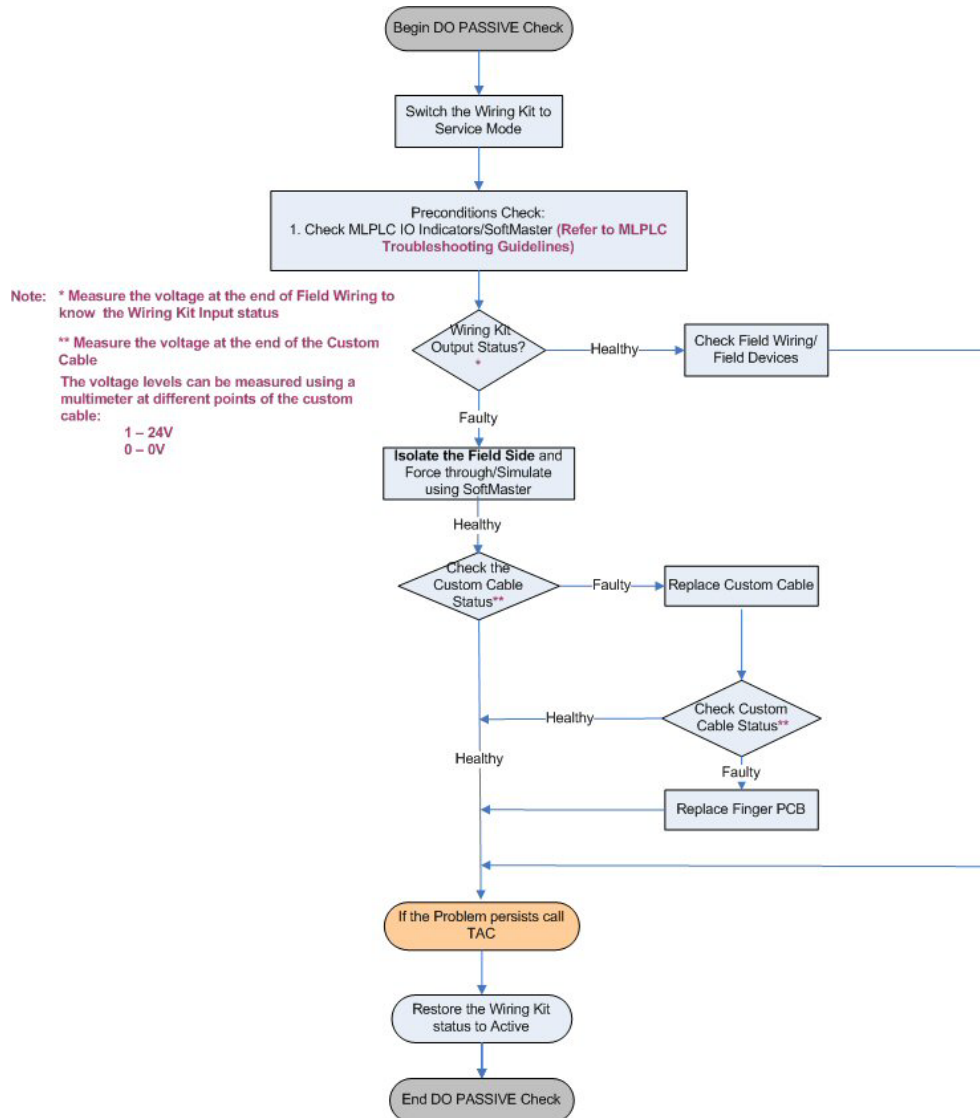
4.2 Maintaining the Wiring Kit

If a failure is detected on one or more DO channels associated with an Active Finger PCB, perform the steps mentioned in the flowchart for troubleshooting.



Note: Always terminate the wiring kit end of the custom cable first.

If a failure is detected on one or more DO channels associated with a Passive Finger PCB, perform the steps mentioned in the flowchart for troubleshooting.

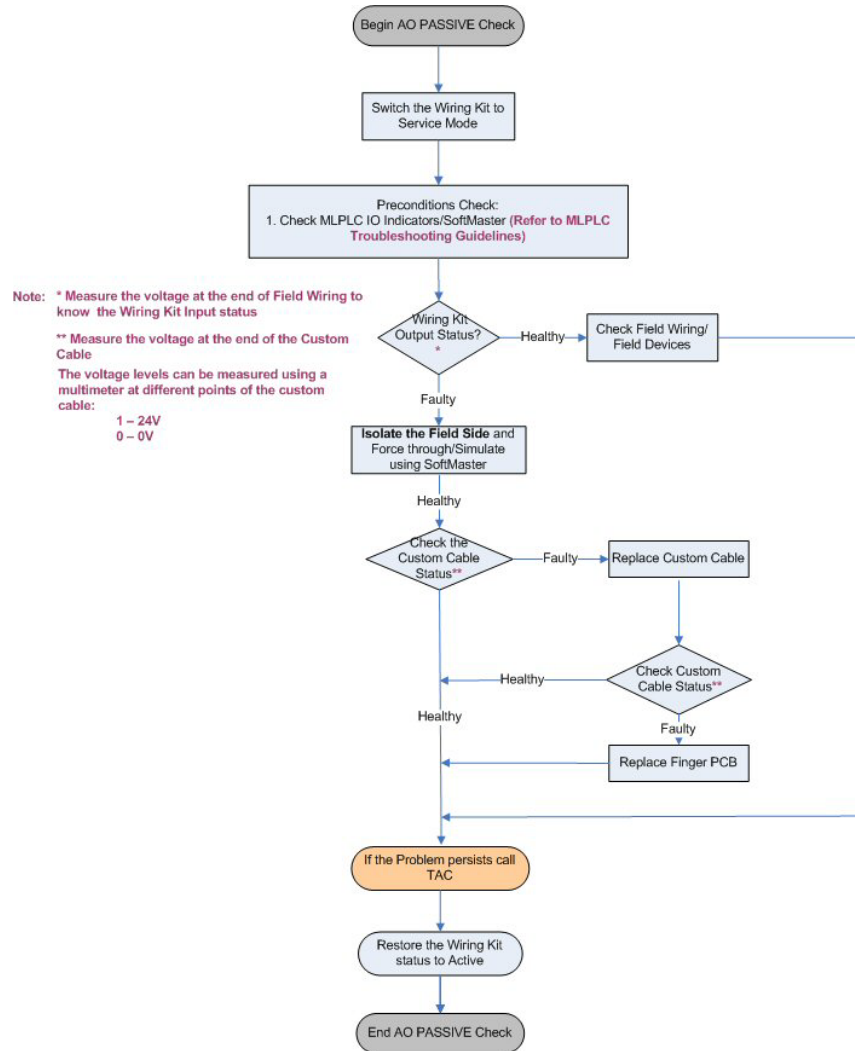


Note: Always terminate the wiring kit end of the custom cable first.

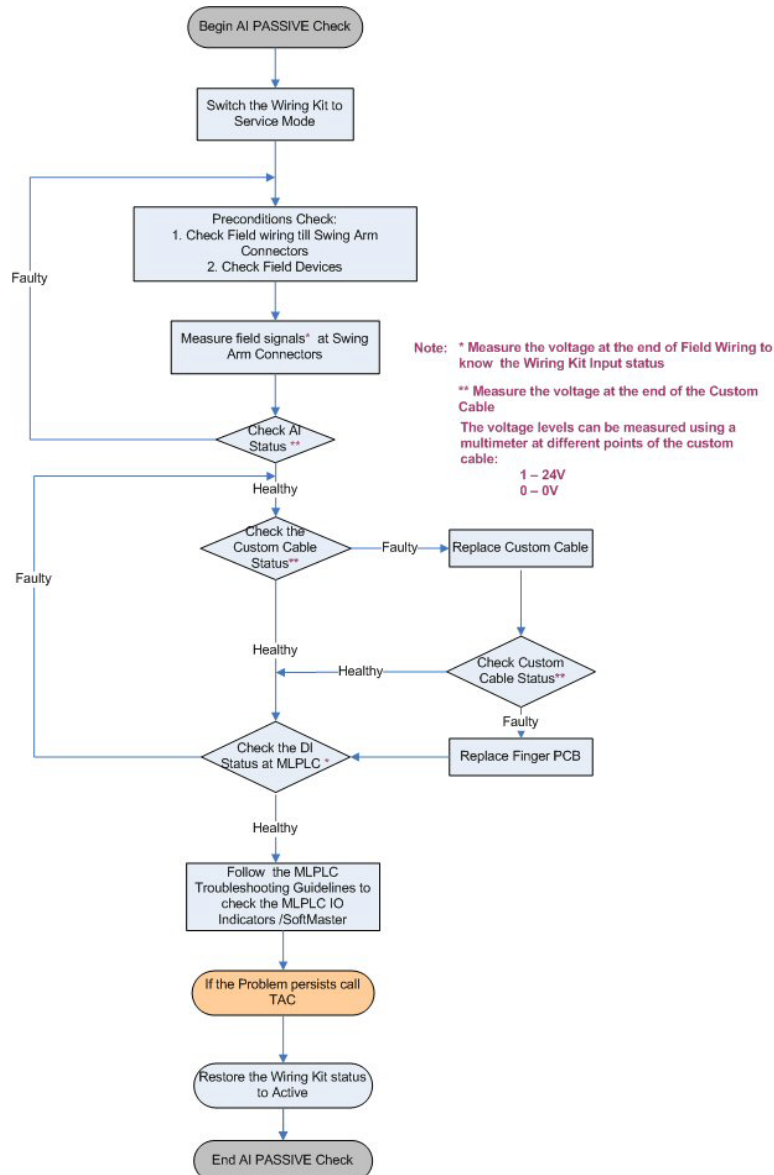
4 Service and maintenance

4.2 Maintaining the Wiring Kit

If a failure is detected on one or more AO channels associated with a Passive Finger PCB, perform the steps mentioned in the flowchart for troubleshooting.



If a failure is detected on one or more AI channels associated with a Passive Finger PCB, perform the steps mentioned in the flowchart for troubleshooting.



4 Service and maintenance

4.2 Maintaining the Wiring Kit



REFERENCE - INTERNAL

For more information about maintaining the MLPLC processor, see the *Programmable Logic Controller MasterLogic-200R User's Guide*.

Removing the finger PCB

The faulty finger PCB must be removed and replaced with the new one.



SHOCK HAZARD

The wiring kit supports modules with a rating of 110VAC or 220VAC. An attempt to directly disconnect a finger PCB connected to these modules might result in an electrical hazard.

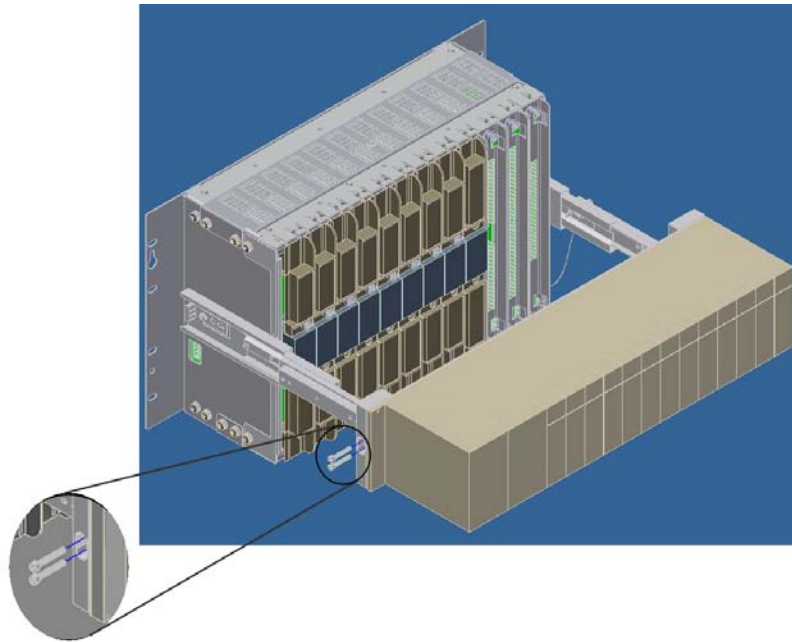
Avoid direct contact with the exposed pins of the finger PCBs.

To avoid any accidents in the plant, ensure to disconnect the finger PCBs by performing the following steps.

Step	Action
1	Ensure that the Wiring Kit is in service mode.
 ESD HAZARD Ensure to open the ESD cover carefully.	
2	Remove the custom cable connected to the MLPLC.
3	Remove the MLPLC I/O module.
4	After disconnecting the I/O module, remove the custom cable connected to the finger PCB in the Wiring Kit.
5	Remove the finger PCB.

Unmounting the MLPLC IO Rack from the Wiring Kit

The specific screws must be removed in sequence, for unmounting the MLPLC I/O rack from the Wiring Kit. The following drawings illustrate the sequence.



4 Service and maintenance

4.2 Maintaining the Wiring Kit

Honeywell

Honeywell Process Solutions
1860 W. Rose Garden Lane
Phoenix, AZ 85027 USA